

Blockchain-Based Lightweight Message Authentication for Edge-Assisted Cross-Domain Industrial Internet of Things

Fengqun Wang[✉], Jie Cui[✉], *Senior Member, IEEE*, Qingyang Zhang[✉], Debiao He[✉], Chengjie Gu[✉],
and Hong Zhong[✉]

Abstract— In edge-assisted cross-domain Industrial Internet of Things (IIoT), blockchain-based authentication is an effective way to build cross-domain trust and secure cross-domain data. However, existing authentication schemes still have serious challenges in terms of efficiency and security. In this article, we propose a blockchain-based lightweight message authentication scheme. First, to address efficiency challenges, we build a blockchain-enabled edge-assisted lightweight authentication framework. This framework uses edge servers to assist smart devices in achieving cross-domain authentication and effectively reduce redundant interactions between entities. Second, to resolve the security challenges, we design a lightweight message authentication algorithm for cross-domain IIoT. The algorithm guarantees message security with low computational overhead and is suitable for multi-receiver cross-domain IIoT. The security proof and analysis demonstrate that the proposed scheme is secure under the random oracle model and can resist various attacks. The performance evaluation shows that our proposed scheme is superior in terms of computation and communication overhead when compared with other related schemes.

Index Terms—Industrial Internet of Things (IIoT), consortium blockchain, cross-domain authentication, elliptic curve cryptography (ECC).

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Fengqun Wang, Jie Cui, Qingyang Zhang, and Hong Zhong are with the Key Laboratory of Intelligent Computing and Signal Processing of Ministry of Education, School of Computer Science and Technology, Anhui University, Hefei 230039, China, also with the Anhui Engineering Laboratory of IoT Security Technologies, Anhui University, Hefei 230039, China, and also with the Institute of Physical Science and Information Technology, Anhui University, Hefei 230039, China (e-mail: fqwang108@foxmail.com; cuijie@mail.ustc.edu.cn; qingyang.zhang.inchina@gmail.com; zhongh@ahu.edu.cn).

Debiao He is with the School of Cyber Science and Engineering, Wuhan University, Wuhan 430072, China (e-mail: hedebiao@163.com).

Chengjie Gu is with Security Research Institute, New H3C Group, Hefei 230088, China (e-mail: gu.chengjie@h3c.com).

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I. INTRODUCTION

IN RECENT years, the introduction of edge computing [1], [2] into the Industrial Internet of Things (IIoT) [3], [4], [5] has shortened decision latency, simplified network topology, and optimized device management in industrial manufacturing [6], [7]. However, as manufacturing becomes more complex, producing a product often requires smart devices from multiple administrative domains (e.g., different smart factories) to collaborate in real time [8], [9].

Fig. 1 shows a typical data exchange scenario in the edge-assisted cross-domain IIoT. In this scenario, multiple smart devices can support the same IIoT service (e.g., the same production task) [10], [11], so this scenario is generally a multi-receiver scenario, i.e., a message has multiple receivers. These devices generate real-time data according to the service type, exchange data with the assistance of edge servers, and process the received data in time. However, the data faces many security issues throughout the transmission process [12], [13]. For example, malicious network attackers can intercept and tamper with real-time data. Once this tampered data is used, it could lead to disruptions in industrial production and economic losses. On the other hand, trust between multiple domains is difficult to build. That is, multiple administrative domains do not trust each other and are reluctant to share sensitive data.

To secure cross-domain data and build trust between multiple administrative domains, many researchers have introduced consortium blockchain technology [14], [15], [16] and proposed cross-domain authentication schemes [17], [18]. However, existing blockchain-based cross-domain authentication schemes face several challenges.

- A blockchain-enabled authentication framework should be lightweight. Most existing blockchain-enabled frameworks have complex organizational structures. For example, the authentication framework is complex in the scheme proposed by Shen et al. [19]. Verifying the legitimacy of a message requires multiple interactions between multiple entities (smart device, authentication agent server, and blockchain). In the multiple-receiver IIoT, these interactions increase as the number of receivers increases. This generates huge communication overheads and non-negligible communication latency, which may eventually

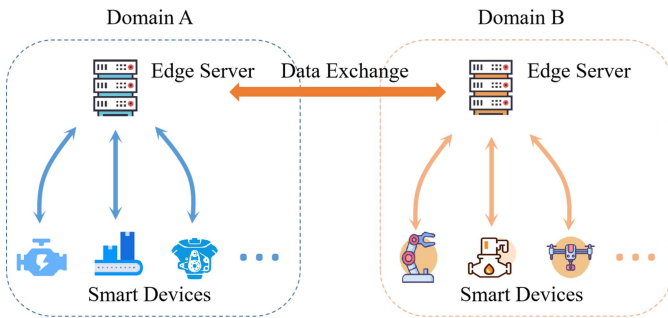


Fig. 1. Data exchange between two domains based on edge computing.

lead to conflicts between collaborating devices and disrupt the production process [20].

- An authentication algorithm should be lightweight. IIoT requires a high level of real-time data. For example, in some industrial control applications, feedback is required within 10 – 100 milliseconds (ms) [21]. Most existing authentication schemes involve many time-consuming cryptographic operations. In this case, smart devices with limited computing power cannot process massive amounts of real-time data in time, leading to significant delays in production decisions [13] and chaotic industrial production. In the multi-receiver IIoT, this challenge becomes more prominent because the time-consuming operation of most existing schemes increases linearly with the number of receivers.
- The security of the authentication algorithm should be more comprehensive. Most existing authentication algorithms do not consider data confidentiality and anonymity simultaneously. If data confidentiality is not guaranteed, attackers can easily obtain industry-critical data (e.g., business secrets), which can lead to significant financial losses [4], [22]. If data anonymity is not guaranteed, attackers can mark the real identity of a smart device and capture the data sent by that device for analysis, leading to a privacy leakage of the device or even the IIoT [23], [24].

To solve the above problems and balance the relationship between real-time and security, we design a blockchain-based lightweight message authentication scheme. In the proposed scheme, we use blockchain to build trust between multiple administrative domains.

The contributions of the proposed scheme are as follows:

- We construct a blockchain-enabled lightweight cross-domain authentication framework. The framework uses edge servers to assist smart devices in cross-domain communication. Compared to existing cross-domain authentication frameworks, the proposed framework involves fewer entities and requires fewer interactions between entities to complete authentication.
- We design a blockchain-based lightweight message authentication algorithm, which is composed of elliptic curve cryptography and hash functions. In addition, we implement our proposed scheme and compare its performance with those of three other related schemes. The theoretical

analysis and simulation results show that our proposed scheme has a lower computational overhead and is superior to the multi-receiver cross-domain IIoT.

- The security proof shows that the proposed scheme satisfies confidentiality and unforgeability. In addition, we conduct a security analysis and compare the security with three other schemes. The results show that our proposed scheme is highly secure, especially considering data confidentiality and anonymity.

The remainder of this paper is organized as follows. Section II presents work related to authentication in IIoT. Section III discusses the preliminary knowledge. Section IV offers the specific details of the framework design. Section V presents details of the proposed scheme. Section VI focuses on the security proof and analysis of the proposed scheme. Section VII provides an experimental evaluation of the proposed scheme. Finally, Section VIII concludes the paper.

II. RELATED WORK

In this section, we mainly introduce and analyze the work related to security authentication in IIoT.

In recent years, with the development of IIoT, the security of IIoT has also received more and more attention from industry and academia. Considering the limited computing power of smart devices in IIoT, Esfahani et al. [25] proposed a lightweight authentication scheme that enables device-to-device authentication. In this scheme, there are only hash and XOR operations, so it has the advantages of low computation and communication overhead. Regarding security, it can resist common attacks such as replay attacks. In 2020, Verma et al. [26] proposed an efficient proxy signature scheme for securing IIoT data, dramatically improving performance. However, neither of these two schemes guarantees the anonymity of messages.

In 2018, Esposito et al. [27] designed a message authentication scheme using group signature. In terms of efficiency, this scheme uses short group signatures to reduce computational overhead. Still, for security, the scheme mainly focuses on the integrity and anonymity of the message and does not consider the confidentiality of messages. Subsequently, Cui et al. [28] used proxy re-encryption techniques to design an anonymous message authentication scheme. This scheme introduces edge computing to effectively reduce messages' transmission delay and guarantees messages' integrity, anonymity, and confidentiality. These two schemes focus on the security of a single administrative domain, and trust between all entities is based on trust in a trusted authority (TA). However, in a cross-domain IIoT environment, each domain does not trust the other. To establish trust between each domain, the most common approach is to have multiple domains trust a specific domain's TA, or to establish a common TA for multiple domains; however, these two approaches are difficult to implement. Therefore, these schemes are not suitable for cross-domain IIoT.

More and more researchers [29], [30], [31], [32] have started using blockchain to solve the above problems. In [29], Guan et al. used the decentralized feature of blockchain to solve the centralized private key generator key escrow problem. The scheme

uses blockchain as an information source for synchronizing user revocation lists, enabling fast detection of user revocations. However, this scheme only achieves data anonymity, not data confidentiality. In [30], Wang et al. used blockchain to achieve efficient conditional anonymity and key management, and the scheme has advantages in terms of communication cost and security. Still, this scheme does not take into account data confidentiality. In 2022, Wang et al. [33] abstracted smart devices in multiple domains into an undirected graph. They designed an authentication scheme using dynamic accumulator and digital signature techniques, which has good performance in terms of efficiency, but the scheme does not specifically consider data anonymity and confidentiality. To achieve both confidentiality and anonymity of data, some researchers have started to combine existing encryption and signature algorithms. For example, Shen et al. [19] designed a secure and practical cross-domain authentication scheme, which combines encryption and group signature technology. The scheme achieves data anonymity and confidentiality. However, in terms of efficiency, the scheme contains many time-consuming cryptographic operations, resulting in high computational and communication overheads. In terms of security, the scheme does not achieve unlinkability of data. In addition, the scheme uses blockchain to solve the trust problem between multiple domains, but the authentication framework is complex, resulting in many interactions between entities during the authentication process. Subsequently, Yang et al. [34] proposed a blockchain-based lightweight authentication scheme that offers significant advantages in terms of computational overhead. The scheme does not meet the data confidentiality. In addition, the above studies do not consider multi-receiver cross-domain IIoT scenarios. They also fail to consider that smart devices in different administrative domains are geographically dispersed and cannot directly cross-domain interaction, because these devices' communication and mobility capabilities are limited.

In summary, although some authentication schemes have been proposed, the existing schemes do not fully consider the complexity of the system framework, the lightweight of the authentication algorithm, the confidentiality and anonymity of the data in the multi-receiver IIoT scenario. Therefore, it is necessary and meaningful to design a secure and efficient authentication scheme for the multi-receiver cross-domain IIoT.

III. PRELIMINARIES

In this section, to better understand our proposed scheme, we introduce the elliptic curve cryptosystem and blockchain used in the proposed scheme.

A. Elliptic Curve Cryptosystem

Let $\mathbb{F}_p$ be a finite field, which is determined by a prime number p . Let a set of elliptic curve point E over $\mathbb{F}_p$ be defined by the equation: $y^2 = x^3 + ax + b \mod p$, where $a, b \in \mathbb{F}_p$. Let O be an infinity point, then O and other points on E make up an additive elliptic curve group $\mathbb{G}$ with the order q and generator P . The main properties of $\mathbb{G}$ are listed below:

- **Scalar point multiplication:** Let $P \in \mathbb{G}$ and $m \in \mathbb{Z}_q^*$, the scalar multiplication of E is defined as $m \cdot P = P + P + \dots + P$ (m times). It is worth noting that the "+" here indicates the point addition.
- **Elliptic Curve Discrete Logarithm problem (ECDLP):** $x \in \mathbb{Z}_q^*$, $Q = xP$, where $P, Q \in \mathbb{G}$ on curve E . Given $Q = xP$, it is computational hard for a probabilistic polynomial-time (PPT) adversary to calculate x .
- **Elliptic Curve Diffie-Hellman Problem (ECDHP):** $x, y \in \mathbb{Z}_q^*$, and $X = xP, Y = yP$, where $X, Y \in \mathbb{G}$ on curve E . Given $X = xP$ and $Y = yP$, it is computational hard for a PPT adversary to calculate xyP .

B. Blockchain

A blockchain is essentially a distributed, shared, and tamper-proof database ledger [35]. Based on the degree of centralization [8], blockchain can be divided into three types: public blockchain, consortium blockchain, and private blockchain. In the proposed scheme, we focus on the following three characteristics of the blockchain:

- **Tamper proofing:** Data stored on the blockchain are tamper-proof and trustworthy.
- **Decentralization:** Blockchain is not dependent on third-party institutions and can maintain databases based on consensus protocols.
- **Smart contract:** A smart contract [36] is a computer program running on the blockchain that can be executed automatically and accurately according to a contract. Once a smart contract is deployed, it will not be modified.

Taking efficiency and security into account, we choose a consortium blockchain that can only be accessed by authorized organizations.

IV. FRAMEWORK DESIGN

In this section, we first describe the blockchain-enabled authentication framework. Then, we present the outline and the security model of the proposed scheme. Finally, we introduce the security objectives.

A. Blockchain-Enabled Authentication Framework

In this subsection, to introduce the blockchain-enabled authentication framework more clearly, we present the entity overview and the communication process, respectively.

1) **Entity Overview:** We consider the scenario of implementing service-based cross-domain authentication in edge-assisted IIoT with the enablement of blockchain technology. As shown in Fig. 2, we divide the blockchain-enabled authentication framework into three layers based on functionality: the end device layer, the management layer, and the blockchain layer.

The end device layer is responsible for data sensing and processing, so it comprises all the smart devices (SD) in each domain.

The management layer is primarily responsible for managing all data in the domain, including data storing, forwarding, and querying. Therefore, this layer contains all the trusted authority

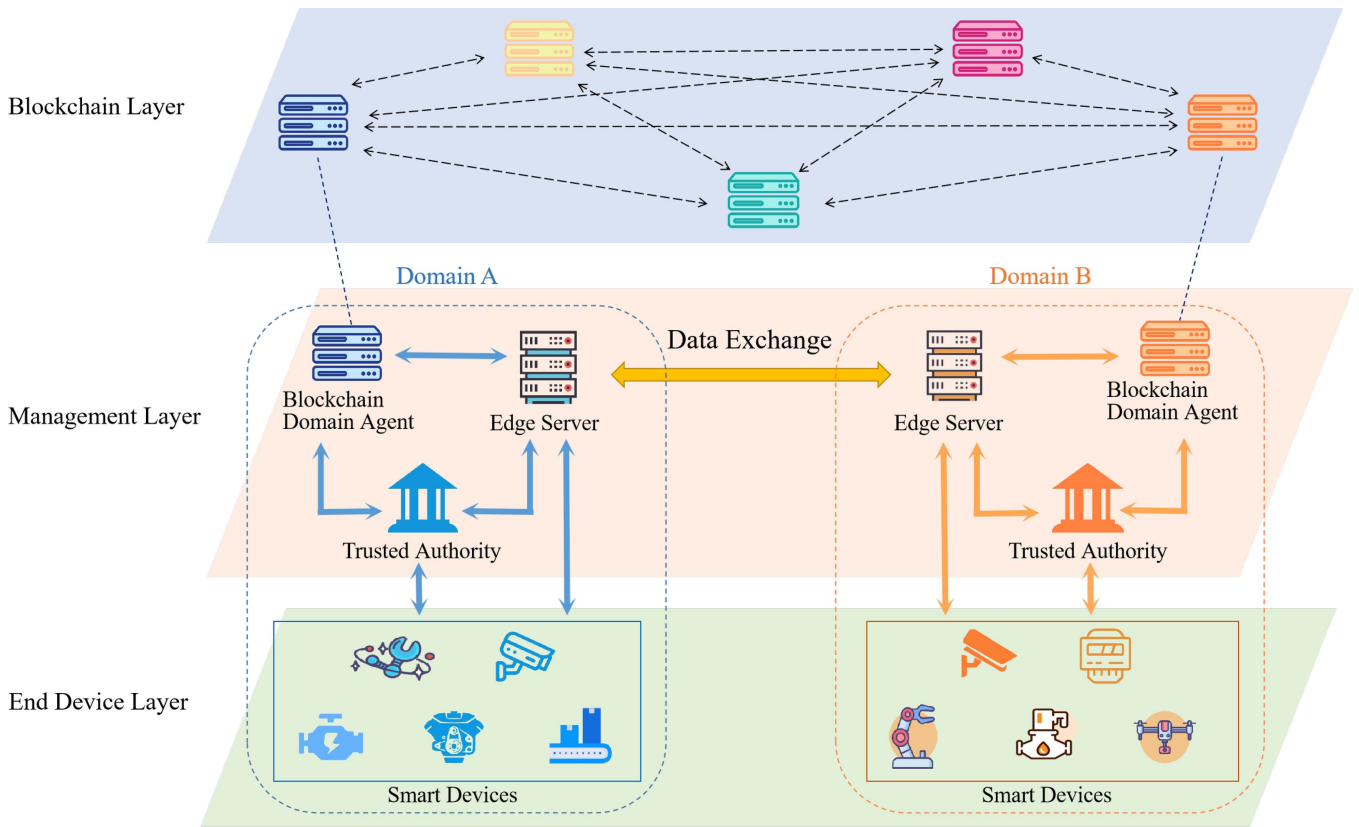


Fig. 2. Blockchain-enabled authentication framework.

(TA), the edge server (ES), and the blockchain domain agents (BCDA) in each domain. The TA, the ES, and the BCDA in this layer collaborate to assist smart devices with secure cross-domain communication and authentication.

The blockchain layer mainly builds trust between multiple administrative domains, and it is composed of BCDAs in the management layer. Specifically, these BCDAs in different domains form a consortium blockchain to maintain a ledger. This ledger contains pseudonym information for each SD, which is tamper-proof and can be used during cross-domain authentication.

The following is a detailed description of the various entities.

- **BCDA:** Each administrative domain has a BCDA, and all BCDAs form a consortium blockchain. The BCDA from different domains can issue smart contracts through negotiation. In each domain, BCDA can collaborate with TA to dynamically update the domain's latest information based on IIoT services and revoke illegal smart devices.
- **TA:** Each administrative domain has a TA. The TA is trusted to all entities in the domain. TA is mainly responsible for generating the security information and managing IIoT services. In addition, TA is the only entity that can trace messages source in its domain.
- **ES:** Each administrative domain has an ES. The ES is primarily responsible for storing and forwarding information from devices, TA, and other administrative domains.

In addition, ES can assist SD in generating service-based keys and cooperate with BCDA to complete necessary data queries. Note that ES as a relay node is not required to be anonymous.

- **SD:** Each administrative domain has numerous SD. They have limited computing, mobility, and communication capabilities. Their main functions are to negotiate service-based keys, sign and encrypt messages to be sent, and decrypt and authenticate messages received. Note that SD's real identity needs to be anonymous in our proposed scheme.

2) **Communication Process:** In IIoT, SDs support different IIoT services. For a cross-domain service, the message sender distributes a piece of data; there may be multiple SDs in the message-receiving domain as receivers of that data. This communication is widely available currently and usually implemented in publish/subscribe systems [37], [38]. However, on the one hand, each domain is relatively independent and geographically dispersed [39], [40]. On the other hand, SDs have limited communication power and mobility. Therefore, SDs distributed in different domains cannot communicate with each other directly. In the blockchain-enabled authentication framework, we provide an ES in each administrative domain to assist SDs in cross-domain communication. That is, when SDs communicate across domains, the data needs to go through the ES.

Our proposed authentication framework is lightweight. For example, in scheme [19], when a smart device receives a message, this smart device needs to send the message to the authentication agent server. Then the authentication agent server needs to query the relevant information in the blockchain. Finally, the authentication agent server verifies the message and sends the verification result to the smart device. However, there is no authentication agent server in the framework we designed. When the message passes through the edge server, the edge server queries the relevant information in the blockchain. Then, the edge server forwards the query result and the message to the smart device. Finally, the smart device directly verifies the received messages. Therefore, our proposed scheme has fewer interactions between entities when performing message verification. In addition, in our proposed scheme, to complete a service-based cross-domain authentication, ES only needs to query the relevant data from the blockchain once, regardless of how many receivers there are. In contrast, in other schemes, the number of queries submitted to the blockchain increases with the number of receivers. Therefore, in a multi-receiver IIoT, our proposed scheme's advantage becomes even more significant.

B. Outline of the Proposed Scheme

The proposed scheme consists of six algorithms, including *Setup*, *Service-Based Key Generation (SBKGen)*, *Sign*, *Encrypt*, *Decrypt* and *Verify*, which are defined as follows.

- 1) *Setup*(1^λ): The algorithm is executed by TA. Given the random system security parameter λ , the TA outputs public system parameters $params$, system secret key msk and system public key P_{pub} .
- 2) *SBKGen*($serv_t, PK_{ES}^B, ID_{ES}^B, PK_{ES}^A, sk_{head}^B$): The algorithm is executed by a smart device SD_{head}^B , which generates the service-based key. Given the service $serv_t$, the public key PK_{ES}^B and identity ID_{ES}^B of the ES in the message receiving domain, the public key PK_{ES}^A of the ES in the message sending domain, and the SD_{head}^B 's secret key sk_{head}^B , it outputs the corresponding $serv_t$ -based secret key sk_{serv_t} and $serv_t$ -based public key PK_{serv_t} . Note that the message sender uses PK_{serv_t} to encrypt the plaintext m to get the ciphertext c , and the message receiver uses sk_{serv_t} to decrypt the ciphertext c to obtain the plaintext m , so we subsequently call the sk_{serv_t} as the decryption key and the PK_{serv_t} as the encryption key.
- 3) *Sign*($sk_{i,j}^A, PID_{i,j}^A, PK_{ES}^B, ID_{ES}^B, PK_{ES}^A, PK_{serv_t}$, m): The algorithm is executed by the message sender SD_i^A . Given $\{PK_{ES}^B, ID_{ES}^B, PK_{ES}^A\}$, the system parameter $params$, signer's secret key $sk_{i,j}^A$, signer's pseudonym $PID_{i,j}^A$, the PK_{serv_t} , and a plaintext m , it outputs some parameters $\{h, u, U\}$ and the corresponding signature σ .
- 4) *Encrypt*($u, U, PK_{serv_t}, PK_{head}^B, h, \sigma, T_i, m$): The algorithm is executed by the message sender SD_i^A . Given the $\{u, U, PK_{serv_t}, PK_{head}^B, h\}$, signature σ , timestamp T_i , and plaintext m , it outputs the ciphertext c .

- 5) *Decrypt*(c, T_i, sk_{serv_t}): The algorithm is executed by the message receiver SD_k^B . Given the ciphertext c , the timestamp T_i , the $serv_t$ -based secret key sk_{serv_t} , it outputs the intermediate parameter U , the corresponding plaintext m and signature σ .
- 6) *Verify*($m, \sigma, U, W, PK_{i,j}^A, sk_{serv_t}, P_{pub}, PID_{i,j}^A, ID_{ES}^B, PK_{ES}^B, PK_{ES}^A$): The algorithm is executed by the message receiver SD_k^B . Given parameters $\{m, \sigma, U, W, PK_{i,j}^A, sk_{serv_t}, P_{pub}, PID_{i,j}^A, ID_{ES}^B, PK_{ES}^B, PK_{ES}^A\}$, it outputs 1 if the input parameters are valid and 0 otherwise.

C. Security Model

In the security model, we divide the adversary into two categories: $\mathcal{A}_I$ and $\mathcal{A}_{II}$. Among them, $\mathcal{A}_I$ is the adversary that attacks the confidentiality of the scheme, and $\mathcal{A}_{II}$ is the adversary that attacks the unforgeability of the scheme.

1) Confidentiality

Definition 1: Our proposed scheme meets confidentiality if the probability that the adversary $\mathcal{A}_I$ could solve the ECDHP is negligible in any polynomial time. In our proposed scheme, confidentiality is defined as indistinguishability under the adaptive-chosen-ciphertext attack (IND-CCA2).

Game 1: The game is an interaction between the simulator $\mathcal{B}$ and the adversary $\mathcal{A}_I$ under IND-CCA2. The specific definition of this game is as follows:

- *Setup:* Given the security parameter λ , the simulator $\mathcal{B}$ outputs the system parameters $params$, system public key P_{pub} , system secret key msk , and then $\mathcal{B}$ sends $(P_{pub}, params)$ to the adversary $\mathcal{A}_I$ and keep msk secret.
- *H₁-query:* Adversary $\mathcal{A}_I$ sends a point $\langle X_i \rangle$ to $\mathcal{B}$, where $X_i \in \mathbb{G}$. The $\mathcal{B}$ returns the corresponding hash value r_{H_1} to $\mathcal{A}_I$, where $r_{H_1} \in \mathbb{Z}_q^*$.
- *H₂-query:* Adversary $\mathcal{A}_I$ sends a point $\langle SPK \rangle$ to $\mathcal{B}$, where $SPK \in \mathbb{G}$. The $\mathcal{B}$ returns the corresponding hash value r_{H_2} to $\mathcal{A}_I$, where $r_{H_2} \in \{0, 1\}^*$.
- *H₃-query:* Adversary $\mathcal{A}_I$ sends $\langle ID_{ES}^B, PK_{ES}^B, PK_{ES}^A \rangle$ to $\mathcal{B}$, the $\mathcal{B}$ returns the corresponding hash value r_{H_3} to $\mathcal{A}_I$, where $r_{H_3} \in \mathbb{Z}_q^*$.
- *H₄-query:* Adversary $\mathcal{A}_I$ sends $\langle K, T_i \rangle$ to $\mathcal{B}$, where $K \in \mathbb{G}$ and T_i is a timestamp. The $\mathcal{B}$ returns the corresponding hash value r_{H_4} to $\mathcal{A}_I$, where $r_{H_4} \in \{0, 1\}^*$.
- *H₅-query:* Adversary $\mathcal{A}_I$ sends a message $\langle m \rangle$ to $\mathcal{B}$, where $m \in \{0, 1\}^*$. The $\mathcal{B}$ returns the corresponding hash value r_{H_5} to $\mathcal{A}_I$, where $r_{H_5} \in \mathbb{Z}_q^*$.
- *Extract encryption key query:* Adversary $\mathcal{A}_I$ sends a service $\langle serv_t \rangle$ to $\mathcal{B}$, and $\mathcal{B}$ returns the encryption key PK_{serv_t} corresponding to the $\langle serv_t \rangle$ to $\mathcal{A}_I$.
- *Extract decryption key query:* Adversary $\mathcal{A}_I$ sends a service $\langle serv_t \rangle$ to $\mathcal{B}$, and $\mathcal{B}$ returns the decryption key sk_{serv_t} corresponding to the $\langle serv_t \rangle$ to $\mathcal{A}_I$.
- *Extract verification key query:* Adversary $\mathcal{A}_I$ sends a pseudonym $\langle PID_{i,j}^A \rangle$ to $\mathcal{B}$, and $\mathcal{B}$ returns the verification key $PK_{i,j}^A$ corresponding to the $\langle PID_{i,j}^A \rangle$ to $\mathcal{A}_I$.
- *Extract signature key query:* Adversary $\mathcal{A}_I$ sends a pseudonym $\langle PID_{i,j}^A \rangle$ to $\mathcal{B}$, and $\mathcal{B}$ returns the signature key $sk_{i,j}^A$ corresponding to the $\langle PID_{i,j}^A \rangle$ to $\mathcal{A}_I$.

- *Encryption key replacement query*: Adversary $\mathcal{A}_I$ sends a service $serv_t$ and a valid new encryption key PK'_{serv_t} to $\mathcal{B}$, $\mathcal{B}$ replaces the original encryption key PK_{serv_t} with the PK'_{serv_t} .
- *Encryption query*: Adversary $\mathcal{A}_I$ sends an encryption query for a message m to $\mathcal{B}$, then the $\mathcal{B}$ returns the corresponding encryption result δ to $\mathcal{A}_I$.
- *Decryption query*: Adversary $\mathcal{A}_I$ sends a decryption query for a message δ to $\mathcal{B}$, then the $\mathcal{B}$ returns the corresponding decryption result m to $\mathcal{A}_I$.

Challenge: After the above queries are over, adversary $\mathcal{A}_I$ sends two plaintext $\{m_0, m_1\}$ to $\mathcal{B}$. The $\mathcal{B}$ randomly chooses a bit $d \in \{0, 1\}$, calculates the encryption message δ corresponding to m_d , and finally returns the δ to $\mathcal{A}_I$. $\mathcal{A}_I$ asks $\mathcal{B}$ some queries. However, the $\mathcal{A}_I$ cannot query for the decryption key of δ and cannot query for the decryption result of δ . Finally, the $\mathcal{A}_I$ outputs a bit $d' \in \{0, 1\}$. If $d' = d$, $\mathcal{A}_I$ wins the game; otherwise, $\mathcal{A}_I$ fails.

2) Unforgeability

Definition 2: Our proposed scheme meets unforgeability if the probability that the adversary $\mathcal{A}_{II}$ could solve the ECDLP is negligible in any polynomial time. In our proposed scheme, the unforgeability is defined as existential unforgeability under chosen message attack (EUF-CMA).

Game 2: The game is an interaction between the simulator $\mathcal{B}$ and the adversary $\mathcal{A}_{II}$ under EUF-CMA. The specific definition of this game is as follows:

- *Setup, Hash query, Extract signature key query, Extract verification key query*: Adversary $\mathcal{A}_{II}$ and $\mathcal{B}$ perform the same operations as in Game 1.
- *Verification key replacement query*: Adversary $\mathcal{A}_{II}$ sends a pseudonym $PID_{i,j}^A$ and a valid new verification key $PK_{i,j}^{A'}$ to $\mathcal{B}$, $\mathcal{B}$ replaces the original verification key $PK_{i,j}^A$ with the $PK_{i,j}^{A'}$.
- *Sign query*: Adversary $\mathcal{A}_{II}$ sends a sign query for a message m to $\mathcal{B}$, then the $\mathcal{B}$ returns the corresponding sign result σ and some necessary parameters to $\mathcal{A}_{II}$.
- *Verify signature query*: Adversary $\mathcal{A}_{II}$ sends a verify signature query for a signature σ to $\mathcal{B}$, then the $\mathcal{B}$ returns the corresponding verification result to $\mathcal{A}_{II}$.

Forgery: After the above queries are over, adversary $\mathcal{A}_{II}$ returns a forged signature pair (m, σ) . If the signature pair is valid, then $\mathcal{A}_{II}$ wins the game; otherwise, $\mathcal{A}_{II}$ fails. Note that in the forgery process, $\mathcal{A}_{II}$ cannot query for the signature key.

D. Security Objectives

To protect the security of the multi-receiver IIoT, the proposed scheme needs to meet the following security objectives.

- 1) *Message confidentiality*: To prevent privacy-sensitive data from being obtained by malicious network attackers, message senders should encrypt messages, and the encrypted messages can only be decrypted by legal message receivers.
- 2) *Message integrity and authentication*: To guarantee the security of the IIoT system, it is necessary to ensure the integrity of the messages and ensure that the message

receiver can verify the messages' validity after receiving them. If a message has been tampered with, the receiver should detect it in time.

- 3) *Message anonymity*: To guarantee the privacy of smart devices, the smart devices' real identity should remain anonymous from malicious network attackers and third parties. Other than the TA in the same domain as the smart device, no other entities can obtain the real identity of the smart device.
- 4) *Un-linkability*: To protect privacy, malicious network attackers and third parties cannot link two messages generated by two pseudonyms.
- 5) *Traceability*: Suppose a message is found to be illegal. In that case, the TA in the same administrative domain as the smart device can extract the pseudonym through messages and then should be able to trace the smart device's real identity corresponding to the pseudonym.
- 6) *Identity revocation*: When a smart device is found to have sent an illegal message, the TA in the same administrative domain should have the ability to revoke the identity of the smart device.
- 7) *Resistance to attacks*: To guarantee the security of IIoT system, the proposed scheme should be able to withstand various common attacks such as the replay attack, the modification attack, and the impersonation attack.

V. PROPOSED SCHEME

In this section, we describe our scheme in detail. As shown in Fig. 3, the proposed scheme can be divided into three main phases: system initialization, service-based key negotiation, message signing and verification. The message signing and verification phase contains two parts: message signing and encryption, message decryption and verification. In addition, the proposed scheme supports illegal message traceability and identity revocation. Therefore, to introduce the proposed scheme more clearly, in this section, we present it in the following five parts: system initialization, service-based key generation, message signing and encryption, message decryption and verification, illegal message tracing and identity revocation. The notations used in this process are shown in Table I.

A. System Initialization

When a new administrative domain X is added to the IIoT system, the blockchain domain agent $BCDA^X$ joins the blockchain according to the strategy configured in the blockchain. Given the public parameters $(p, q, E, \mathbb{G}, \mathbb{Z}_q^*)$ for administrative domain X , then TA^X initializes the domain X as per the following steps.

- 1) The TA^X selects a random number $msk^X \in \mathbb{Z}_q^*$ as the system master secret key in domain X and computes the corresponding public key $P_{pub}^X = msk^X \cdot P$.
- 2) TA from different domains negotiate several secure one-way hash functions: $H_1 : \mathbb{G} \rightarrow \mathbb{Z}_q^*$, $H_2 : \mathbb{G} \rightarrow \{0, 1\}^*$, $H_3 : \{0, 1\}^* \times \mathbb{G} \times \mathbb{G} \rightarrow \mathbb{Z}_q^*$, $H_4 : \mathbb{G} \times \{0, 1\}^* \rightarrow \{0, 1\}^*$, $H_5 : \{0, 1\}^* \rightarrow \mathbb{Z}_q^*$. Then, the system parameters $params$ will be published in IIoT, which include $\{q, \mathbb{G}, E, P, P_{pub}^X, \mathbb{Z}_q^*, H_1, H_2, H_3, H_4, H_5\}$.

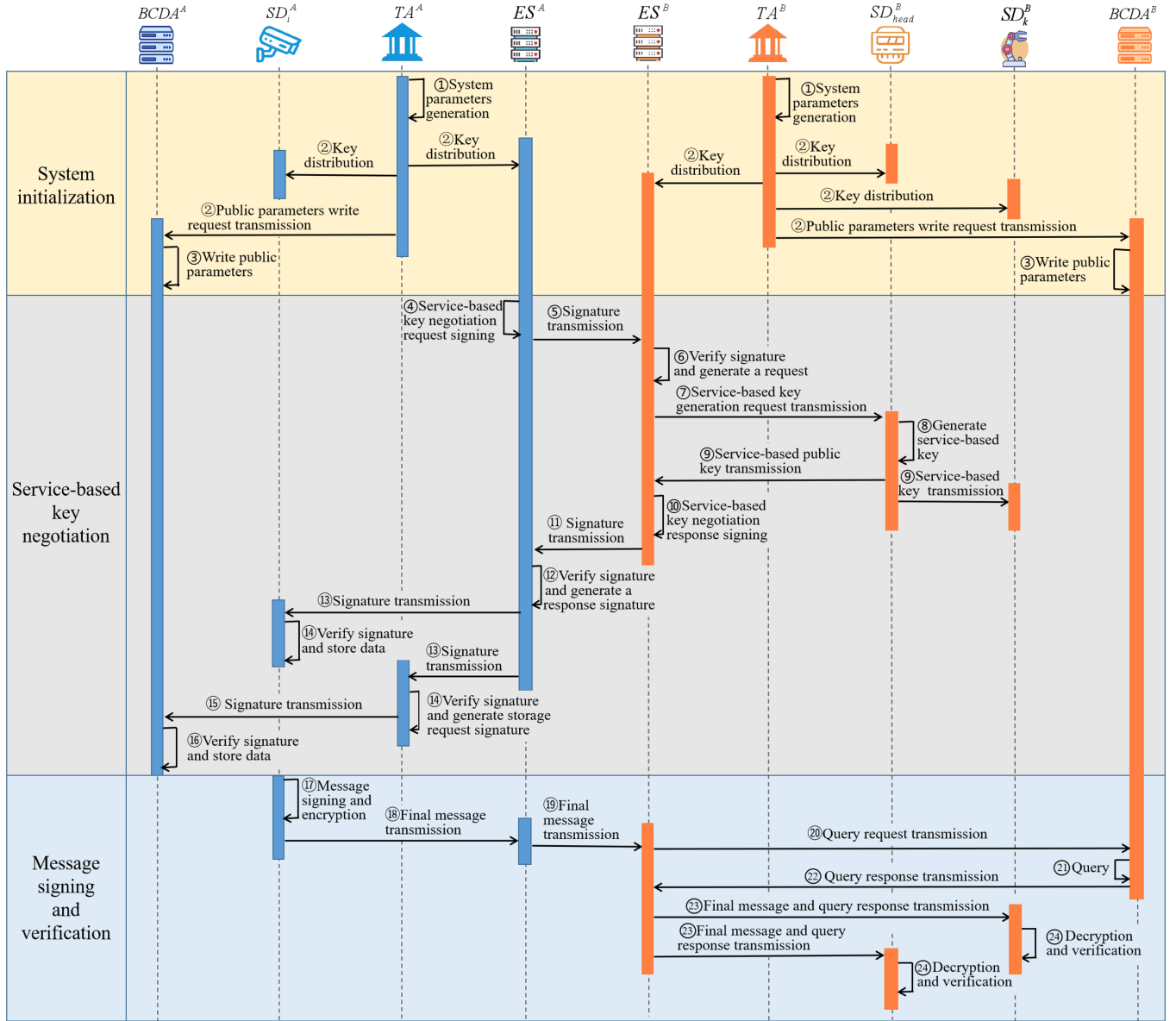


Fig. 3. Message authentication process overview.

- 3) TA^X selects a random number $sk_{ES}^X \in \mathbb{Z}_q^*$ as the secret key for ES^X and computes the corresponding public key $PK_{ES}^X = sk_{ES}^X \cdot P$. Then, the TA^X sends $\{sk_{ES}^X, PK_{ES}^X\}$ to ES^X via a secure channel.
- 4) When a new smart device is added to administrative domain X, the TA^X first selects a real identity RID_i^X for the smart device SD_i^X . Then, the TA^X selects a random number $r_j \in \mathbb{Z}_q^*$ and computes $sk_{i,j}^X = H_1(r_j \cdot P_{pub}) + msk^X$ as a secret key for SD_i^X . To ensure the anonymity of a message, TA^X computes pseudonym $PID_{i,j}^X = RID_i^X \oplus H_2(sk_{i,j}^X \cdot P_{pub}^X)$ and calculates public key $PK_{i,j}^X = sk_{i,j}^X \cdot P$ for SD_i^X . Subsequently, the TA^X sends $\{sk_{i,j}^X, PK_{i,j}^X, PID_{i,j}^X, VP_{i,j}^X\}$ to SD_i^X via a secure channel, where $VP_{i,j}^X$ indicates the validity period of the $PID_{i,j}^X$. Noting that $sk_{i,j}^X$ and

- $PID_{i,j}^X$ are updated every period of time, so TA^X will generate multiple set of data $(sk_{i,1}^X, PK_{i,1}^X, PID_{i,1}^X, VP_{i,1}^X), (sk_{i,2}^X, PK_{i,2}^X, PID_{i,2}^X, VP_{i,2}^X), (sk_{i,3}^X, PK_{i,3}^X, PID_{i,3}^X, VP_{i,3}^X), \dots, (sk_{i,n}^X, PK_{i,n}^X, PID_{i,n}^X, VP_{i,n}^X)$ to SD_i^X .
- 5) TA^X sends administrative domain X initial parameters $\{P_{pub}^X, PK_{ES}^X\}$ to $BCDA^X$, then the $BCDA^X$ triggers a smart contract to write these initial parameters into the blockchain.

B. Service-Based Key Negotiation

Suppose the message sender is the entity in domain A, and the receiver is in domain B. Given a new service (e.g., $serv_t$), entities from domains A and domain B need to negotiate a $serv_t$ -based key before exchanging information. In our proposed scheme, the

TABLE I
NOTATIONS AND DEFINITIONS USED

Notations	Definitions
$BCDA^X$	Blockchain domain agent in domain X
TA^X	Trusted authority in domain X
ES^X	Edge server in domain X
SD_i^X	i -th smart device in domain X
RID_i^X	Real identity of SD_i^X
$PID_{i,j}^X$	j -th pseudonym of SD_i^X
$VP_{i,j}^X$	Validity period of $PID_{i,j}^X$
msk^X	Master secret key of TA^X
P_{pub}^X	Master public key of TA^X
sk_{ES}^X	Secret key of ES^X
PK_{ES}^X	Public key of ES^X
$sk_{i,j}^X$	j -th secret key of SD_i^X
$PK_{i,j}^X$	j -th public key of SD_i^X
$serv_t$	t -th service
sk_{serv_t}	$serv_t$ -based secret key
PK_{serv_t}	$serv_t$ -based public key
$Sig(\cdot)$	ECDSA signature algorithm
$Ver(\cdot)$	Verify the ECDSA signature
m	Plaintext
c	Ciphertext
σ	Signature
T_i	Timestamp
H_1, H_2, H_3, H_4, H_5	Five secure hash functions

blockchain creates a list for each service, and these lists are used to query information about valid smart devices corresponding to services. For example, if the service is $serv_t$, the corresponding list is $List_{serv_t}$. And the $List_{serv_t}$ stores all valid pseudonyms, the public key, and the validity period of all smart devices that support the service $serv_t$.

It is worth noting that there are multiple message receivers for the same service. However, in the service-based key negotiation process, the $serv_t$ -based public key PK_{serv_t} and the $serv_t$ -based secret key sk_{serv_t} are generated by a certain smart device, which has relatively strong computing power and low computing density among all message receivers. Assume that the smart device SD_{head}^B is the generator of PK_{serv_t} and sk_{serv_t} . Negotiating the $serv_t$ -based key as per the following steps.

- 1) The edge server ES^A sends a request $Sig_{sk_{ES}^A}(serv_t)$ for key negotiation to edge server ES^B .
- 2) Upon receiving the request $Sig_{sk_{ES}^A}(serv_t)$, the ES^B verifies the validity of $Sig_{sk_{ES}^A}(serv_t)$ by computing $Ver_{PK_{ES}^A}(Sig_{sk_{ES}^A}(serv_t))$. If passed, ES^B sends $serv_t$ to the corresponding smart device SD_{head}^B .
- 3) Upon receiving $serv_t$, SD_{head}^B selects a random number $d \in \mathbb{Z}_q^*$ and computes a $serv_t$ -based public key $PK_{serv_t} = d \cdot P$. Then the smart device SD_{head}^B uses its secret key sk_{head} to calculate $serv_t$ -based secret key $sk_{serv_t} = d + sk_{head}^B \cdot h$, where

$h = H_3(ID_{ES}^B, PK_{ES}^B, PK_{ES}^A)$. Subsequently, SD_{head}^B sends PK_{serv_t} to ES^B and sends $\{sk_{serv_t}, PK_{serv_t}\}$ to other smart devices in administrative domain B that support the same service via a secure channel.

- 4) Upon receiving PK_{serv_t} , the edge server ES^B generates signature $Sig_{sk_{ES}^B}(PK_{serv_t}, PK_{head}^B)$ and sends it to ES^A .
- 5) Once ES^A receives the signature $Sig_{sk_{ES}^B}(PK_{serv_t}, PK_{head}^B)$, it first verifies the validity of the signature by computing $Ver_{PK_{ES}^B}(Sig_{sk_{ES}^B}(PK_{serv_t}, PK_{head}^B))$. If passed, ES^A sends $Sig_{sk_{ES}^A}(ID_{ES}^B, PK_{ES}^B, PK_{serv_t}, PK_{head}^B)$ to corresponding smart devices and trusted authority TA^A in administrative domain A.
- 6) Once SD receives $Sig_{sk_{ES}^A}(ID_{ES}^B, PK_{ES}^B, PK_{serv_t}, PK_{head}^B)$, it first verifies the legitimacy of the signature, if it is not legitimate, directly discard; otherwise, store the corresponding $serv_t$ -based data.
- 7) The TA^A submits a signature $Sig_{msk^A}(Store_req, serv_t, PIDS, SDPKS, VPS)$ to $BCDA^A$, where $Store_req$ indicates a storage request, $PIDS$ indicates a set of all valid pseudonyms based on $serv_t$, $SDPKS$ denotes the public key set corresponding to $PIDS$, VPS indicates the validity period set corresponding to $PIDS$. When the $BCDA^A$ receives the signature, it verifies whether the signature is valid. If it is invalid, $BCDA^A$ discards the request directly; otherwise, $BCDA^A$ stores $\{PIDS, SDPKS, VPS\}$ into $List_{serv_t}$.

C. Message Signing and Encryption

Assume that the smart device SD_i^A in administrative domain A is the message sender and the smart device SD_k^B in administrative domain B is the message receiver. To ensure the security of messages, SD_i^A needs to sign and encrypt the message before sending them, as explained below.

- 1) The message sender SD_i^A selects two random numbers $w, u \in \mathbb{Z}_q^*$, and computes $W = w \cdot P$, $U = u \cdot P$ respectively. Then the SD_i^A calculates $h_3 = H_3(PID_{i,j}^A, W, U)$ and $h = H_3(ID_{ES}^B, PK_{ES}^B, PK_{ES}^A)$.
- 2) When a smart device SD_i^A wants to send a plaintext m , it first selects current timestamp T_i and computes signature $\sigma = sk_{i,j}^A \cdot (H_5(m) \cdot w + (h + h_3) \cdot u)^{-1}$. Then it calculates ciphertext $c = (s, Y_i^A)$, where $s = H_4(U, T_i) \oplus (m || \sigma)$, $Y_i^A = u(PK_{serv_t} + h \cdot PK_{head}^B)$.
- 3) SD_i^A sends the final message $\delta = (c, W, PID_{i,j}^A, T_i)$ to SD_k^B through ES^A and ES^B .

Remark 1: In the message signing and encryption process, some data (e.g., $w, u, W, U, h_3, h, Y_i^A$) generation does not require online messages m . Therefore, when the smart device is idle or computational density is not high, the smart device can generate these data in advance through pre-processing and store them for generating online signatures in the future. Using this pre-processing approach can effectively improve the signing efficiency of the smart device.

D. Message Decryption and Verification

Upon receiving a message $\delta' = (c', W', PID_{i,j}^{A'}, T'_i)$, ES^B queries the blockchain to check if the $PID_{i,j}^{A'}$ is valid, and if so, forwards the corresponding data to SD_k^B . SD_k^B first decrypts the ciphertext c' by using sk_{serv_t} and then verifies the integrity of the received signature σ' . The main process is divided into the following steps.

- 1) When ES^B receives the final message δ' , it first queries the blockchain whether the $PID_{i,j}^{A'}$ exists in $List_{serv_t}$ and has not expired. If the result of the query is that the $PID_{i,j}^{A'}$ does not exist or has expired, ES^B will directly discard the final message; otherwise, ES^B will obtain the $PK_{i,j}^{A'}$ corresponding to the $PID_{i,j}^{A'}$ from the blockchain. Finally, ES^B sends the δ' and $PK_{i,j}^{A'}$ to the corresponding smart device SD_k^B .
- 2) Upon receiving this message δ' , SD_k^B checks the freshness of T'_i . Assume that the time of receiving message is T . If $\Delta T \geq T - T'_i$, SD_k^B continues; otherwise, SD_k^B discards the message δ' .
- 3) To obtain plaintext m and signature σ , SD_k^B computes $U' = Y_i^{A'} \cdot sk_{serv_t}^{-1}$, and then calculates $m' || \sigma' = H_4(U', T_i) \oplus s'$.
- 4) To verify the validity and integrity of signature σ' , SD_k^B computes $h'_3 = H_3(PID_{i,j}^{A'}, W', U')$ and then calculates $h = H_3(ID_{ES}^B, PK_{ES}^B, PK_{ES}^A)$. Subsequently, SD_k^B checks whether the formula $PK_{i,j}^{A'} = \sigma' \cdot (H_5(m') \cdot W' + (h + h'_3) \cdot U')$ holds true or not. If not, SD_k^B rejects the δ' ; otherwise, the δ' be considered legal.

Due to $PK_{serv_t} = d \cdot P$, $sk_{serv_t} = d + sk_{head}^B \cdot h$, $W = w \cdot P$, $PK_{i,j}^A = sk_{i,j}^A \cdot P$, $U = u \cdot P$, the correctness of the U' can be ensured using the below formula.

$$\begin{aligned}
 U' &= Y_i^{A'} \cdot sk_{serv_t}^{-1} \\
 &= u(PK_{serv_t} + h \cdot PK_{head}^B) \cdot sk_{serv_t}^{-1} \\
 &= u(d \cdot P + sk_{serv_t} \cdot P - d \cdot P) \cdot sk_{serv_t}^{-1} \\
 &= (u \cdot sk_{serv_t} \cdot P) \cdot sk_{serv_t}^{-1} \\
 &= u \cdot P \\
 &= U
 \end{aligned} \tag{1}$$

The correctness of the verification can be ensured using the below formula.

$$\begin{aligned}
 &\sigma' \cdot (W' \cdot H_5(m') + (h + h'_3) \cdot U') \\
 &= sk_{i,j}^A \cdot (w \cdot H_5(m) + (h + h_3) \cdot u)^{-1} \\
 &\quad \cdot (W \cdot H_5(m) + (h + h_3) \cdot U) \\
 &= sk_{i,j}^A \cdot (w \cdot H_5(m) + (h + h_3) \cdot u)^{-1} \\
 &\quad \cdot (w \cdot H_5(m) + (h + h_3) \cdot u) \cdot P \\
 &= sk_{i,j}^A \cdot P \\
 &= PK_{i,j}^A
 \end{aligned} \tag{2}$$

Remark 2: The pre-processing approach can also improve message decryption and verification efficiency like the message signing and encryption process. When the message has not yet arrived, and the smart device is idle or computational density is not high, some data (e.g., h) can be pre-processed offline and stored for future use in decrypting and verifying received messages.

E. Illegal Message Tracing and Identity Revocation

In the proposed scheme, if it is found that a smart device in the administrative domain X has published an illegal message, then TA^X traces the source of the illegal messages and revokes the identity of the smart device as per the following steps.

- 1) TA^X queries the $PK_{i,j}^X$ corresponding to the $PID_{i,j}^X$, and then computes $temp = msk^X \cdot PK_{i,j}^X$, $RID_i^X = PID_{i,j}^X \oplus H_2(temp)$.
- 2) TA^X stops regenerating the pseudonym $PID_{i,j}^X$ and secret key $sk_{i,j}^X$ corresponding to the RID_i^X . Then the TA^X submits a signature $Sig_{msk^X}(Del_req, PIDS_i^X)$ to the $BCDA^X$ to delete the $PIDS_i^X$, where Del_req indicates delete request and $PIDS_i^X$ indicates a set of all pseudonyms corresponding to the RID_i^X .
- 3) Upon receiving the signature $Sig_{msk^X}(Del_req, PIDS_i^X)$, $BCDA^X$ first verifies the validity of the signature by computing $Ver_{P_{pub}^X}(Sig_{msk^X}(Del_req, PIDS_i^X))$. If the signature is valid, $BCDA^X$ will delete the $\{PIDS_i^X, PKS_i^X, VPS_i^X\}$ in corresponding list, where PKS_i^X denotes the public key set corresponding to $PIDS_i^X$, VPS_i^X indicates the validity period set corresponding to $PIDS_i^X$.

VI. SECURITY PROOF AND ANALYSIS

In this section, we first demonstrate that our proposed scheme is secure through security proof. Then we show that our proposed scheme can resist various common attacks through security analysis.

A. Security Proof

This subsection will perform detailed security proof to prove the security of our proposed scheme.

1) Confidentiality

Theorem 1: In the random oracle model, if an adversary $\mathcal{A}_I$ with probabilistic polynomial time executes *Game 1* and wins the game with a non-negligible probability ε_I , then the simulator $\mathcal{B}$ with probabilistic polynomial time can solve the ECDHP problem with a non-negligible probability no less than $(1 - \frac{q_{dsk}}{2^e}) \frac{\varepsilon_I}{e(q_{enc} + q_{dsk})}$, where q_{dsk} denotes the maximum times of extracting decryption key queries, q_{enc} denotes the maximum times of encryption queries, λ denotes secure parameter, e denotes natural logarithm base.

Proof: If there is an adversary $\mathcal{A}_I$ that can break the proposed scheme with a non-negligible probability ε_I , then we can construct a simulator $\mathcal{B}$ based on $\mathcal{A}_I$, and the $\mathcal{B}$ can solve the ECDHP run by $\mathcal{A}_I$ as a subroutine with non-negligible probability. Given

a group $\mathbb{G}$ and an ECDHP instance $\{P, aP, bP | a, b \in \mathbb{Z}_q^*\}$, $\mathcal{B}$ simulates oracles queried by $\mathcal{A}_I$ as follows.

Setup: Upon receiving the setup query from $\mathcal{A}_I$, the $\mathcal{B}$ sends the public parameters $\{E, \mathbb{G}, P, P_{pub}, H_i (i = 1, 2, 3, 4, 5)\}$ to $\mathcal{A}_I$.

H₁ -Query: For the query, $\mathcal{B}$ presets a map Map_{H1} , and the Map_{H1} is empty at the beginning. When the adversary $\mathcal{A}_I$ makes an H_1 query with $\langle X_i \rangle$, where X_i denotes an elliptic curve point, $\mathcal{B}$ checks whether the Map_{H1} has the key $\langle X_i \rangle$. If so, $\mathcal{B}$ finds the corresponding value and returns it to $\mathcal{A}_I$. Otherwise, $\mathcal{B}$ chooses a random number $r_{H1} \in \mathbb{Z}_q^*$, and r_{H1} should satisfy $r_{H1} \notin Map_{H1}$. Then the $\mathcal{B}$ sets the value $Map_{H1}(\langle X_i \rangle) = r_{H1}$, and returns r_{H1} to $\mathcal{A}_I$.

H₂ -Query: For the query, $\mathcal{B}$ presets a map Map_{H2} , and the Map_{H2} is empty at the beginning. When the adversary $\mathcal{A}_I$ makes an H_2 query with $\langle SPK \rangle$, where SPK denotes an elliptic curve point, $\mathcal{B}$ checks whether the Map_{H2} has the key $\langle SPK \rangle$. If so, $\mathcal{B}$ finds the corresponding value and returns it to $\mathcal{A}_I$. Otherwise, $\mathcal{B}$ chooses a random bit-string $r_{H2} \in \{0, 1\}^*$, and r_{H2} should satisfy $r_{H2} \notin Map_{H2}$. Then the $\mathcal{B}$ sets the value $Map_{H2}(\langle SPK \rangle) = r_{H2}$, and returns r_{H2} to $\mathcal{A}_I$.

H₃ -Query: For the query, $\mathcal{B}$ presets a map Map_{H3} , and the Map_{H3} is empty at the beginning. When the adversary $\mathcal{A}_I$ makes an H_3 query with $\langle ID_{ES}^B, PK_{ES}^B, PK_{ES}^A \rangle$, $\mathcal{B}$ checks whether the Map_{H3} has the key $\langle ID_{ES}^B, PK_{ES}^B, PK_{ES}^A \rangle$. If so, $\mathcal{B}$ finds the corresponding value and returns it to $\mathcal{A}_I$. Otherwise, $\mathcal{B}$ chooses a random number $r_{H3} \in \mathbb{Z}_q^*$, and r_{H3} should satisfy $r_{H3} \notin Map_{H3}$. Then the $\mathcal{B}$ sets the value $Map_{H3}(\langle ID_{ES}^B, PK_{ES}^B, PK_{ES}^A \rangle) = r_{H3}$, and returns r_{H3} to $\mathcal{A}_I$.

H₄ -Query: For the query, $\mathcal{B}$ presets a map Map_{H4} , and the Map_{H4} is empty at the beginning. When the adversary $\mathcal{A}_I$ makes an H_4 query with $\langle K, T_i \rangle$, $\mathcal{B}$ checks whether the Map_{H4} has the key $\langle K, T_i \rangle$. If so, $\mathcal{B}$ finds the corresponding value and returns it to $\mathcal{A}_I$. Otherwise, $\mathcal{B}$ chooses a random bit-string $r_{H4} \in \{0, 1\}^*$, and r_{H4} should satisfy $r_{H4} \notin Map_{H4}$. Then the $\mathcal{B}$ sets the value $Map_{H4}(\langle K, T_i \rangle) = r_{H4}$, and returns r_{H4} to $\mathcal{A}_I$.

H₅ -Query: For the query, $\mathcal{B}$ presets a map Map_{H5} , and the Map_{H5} is empty at the beginning. When the adversary $\mathcal{A}_I$ makes an H_5 query with $\langle m \rangle$, where $m \in \{0, 1\}^*$, $\mathcal{B}$ checks whether the Map_{H5} has the key $\langle m \rangle$. If so, $\mathcal{B}$ finds the corresponding value and returns it to $\mathcal{A}_I$. Otherwise, $\mathcal{B}$ chooses a random number $r_{H5} \in \mathbb{Z}_q^*$, and r_{H5} should satisfy $r_{H5} \notin Map_{H5}$. Then the $\mathcal{B}$ sets the value $Map_{H5}(\langle m \rangle) = r_{H5}$, and returns r_{H5} to $\mathcal{A}_I$.

Extract Encryption Key Query: For the query, $\mathcal{B}$ presets a map Map_{EK} , and the Map_{EK} is empty at the beginning. When the adversary $\mathcal{A}_I$ makes an extract encryption key query with a service $\langle serv_t \rangle$, $\mathcal{B}$ checks whether the Map_{EK} has the key $\langle serv_t \rangle$. If so, $\mathcal{B}$ finds the corresponding value and returns it to $\mathcal{A}_I$. Otherwise, the $\mathcal{B}$ chooses a random number $c_t \leftarrow \{0, 1\}$, and $Pr[c_t = 1] = \frac{1}{q_{enc} + q_{dsk} + 1}$, where the "1" in the denominator denotes that $\mathcal{A}_I$ has selected one service as the challenge service. If $c_t = 0$, the $\mathcal{B}$ chooses two random numbers $sk_{serv_t}, r_H \in \mathbb{Z}_q^*$, then the $\mathcal{B}$ computes $PK_{serv_t} = sk_{serv_t} \cdot P - r_H \cdot PK_{head}$, where the PK_{serv_t} should satisfy

$PK_{serv_t} \notin Map_{EK}$. Final the $\mathcal{B}$ sets $Map_{EK}(\langle serv_t \rangle) = PK_{serv_t}$, and returns PK_{serv_t} to $\mathcal{A}_I$. At the same time, the $\mathcal{B}$ sets $Map_{dsk}(\langle serv_t \rangle) = sk_{serv_t}$. If $c_t = 1$, let $PK_{serv_t} = r_{know} \cdot P$, where the $r_{know} \in \mathbb{Z}_q^*$ that $\mathcal{B}$ already knows. The PK_{serv_t} should satisfy $PK_{serv_t} \notin Map_{EK}$. Otherwise, $\mathcal{B}$ re-chooses r_{know} , then $\mathcal{B}$ sets the value $Map_{EK}(\langle serv_t \rangle) = PK_{serv_t}$, and final returns PK_{serv_t} to $\mathcal{A}_I$.

Extract Decryption Key Query: For the query, $\mathcal{B}$ presets a map Map_{dsk} , and the Map_{dsk} is empty at the beginning. When the adversary $\mathcal{A}_I$ makes an extract decryption key query with a service $\langle serv_t \rangle$, $\mathcal{B}$ checks whether the Map_{dsk} has the key $\langle serv_t \rangle$. If so, $\mathcal{B}$ finds the corresponding value and returns it to $\mathcal{A}_I$. Otherwise, the $\mathcal{B}$ first conducts extract encryption key query and obtains the corresponding response $\langle PK_{serv_t} \rangle$. If $c_t = 0$, it means that $\mathcal{B}$ has been set $Map_{dsk}(\langle serv_t \rangle) = sk_{serv_t}$, and then the $\mathcal{B}$ returns the corresponding sk_{serv_t} to $\mathcal{A}_I$. If $c_t = 1$, $\mathcal{B}$ will abort the process.

Extract Verification Key Query: For the query, $\mathcal{B}$ presets a map Map_{VK} , and the Map_{VK} is empty at the beginning. When the adversary $\mathcal{A}_I$ makes an extract verification key query with a pseudonym $\langle PID_{i,j}^A \rangle$, $\mathcal{B}$ checks whether the Map_{VK} has the key $\langle PID_{i,j}^A \rangle$. If so, $\mathcal{B}$ finds the corresponding value and returns it to $\mathcal{A}_I$. Otherwise, the $\mathcal{B}$ chooses a random number $g_{i,j} \leftarrow \{0, 1\}$, and $Pr[g_{i,j} = 1] = \frac{1}{q_s + q_{ssk} + 1}$, where the "1" in the denominator denotes that $\mathcal{A}_I$ has selected one pseudonym as the challenge pseudonym, q_{ssk} denotes the maximum times of extracting signature key queries, q_s denotes the maximum times of sign queries. If $g_{i,j} = 0$, the $\mathcal{B}$ chooses a random numbers $sk_i \in \mathbb{Z}_q^*$, then the $\mathcal{B}$ computes $PK_{i,j}^A = sk_{i,j} \cdot P$, where the $PK_{i,j}^A$ should satisfy $PK_{i,j}^A \notin Map_{VK}$. Final the $\mathcal{B}$ sets $Map_{VK}(\langle PID_{i,j}^A \rangle) = PK_{i,j}^A$, and returns $PK_{i,j}^A$ to $\mathcal{A}_I$. At the same time, the $\mathcal{B}$ sets $Map_{ssk}(\langle PID_{i,j}^A \rangle) = sk_i$. If $g_{i,j} = 1$, let $PK_{i,j}^A = r_{know2} \cdot P$, where the $r_{know2} \in \mathbb{Z}_q^*$ that $\mathcal{B}$ already knows. The $PK_{i,j}^A$ should satisfy $PK_{i,j}^A \notin Map_{VK}$. Otherwise, $\mathcal{B}$ re-chooses r_{know2} , then $\mathcal{B}$ sets the value $Map_{VK}(\langle PID_{i,j}^A \rangle) = PK_{i,j}^A$, and final returns $PK_{i,j}^A$ to $\mathcal{A}_I$.

Extract Signature Key Query: For the query, $\mathcal{B}$ presets a map Map_{ssk} , and the Map_{ssk} is empty at the beginning. When the adversary $\mathcal{A}_I$ makes an extract signature key query with a pseudonym $\langle PID_{i,j}^A \rangle$, $\mathcal{B}$ checks whether the Map_{ssk} has the key $\langle PID_{i,j}^A \rangle$. If so, $\mathcal{B}$ finds the corresponding value and returns it to $\mathcal{A}_I$. Otherwise, the $\mathcal{B}$ first conducts extract verification key query and obtains the corresponding response $\langle PK_{i,j}^A \rangle$. If $g_{i,j} = 0$, it means that $\mathcal{B}$ has been set $Map_{ssk}(\langle PID_{i,j}^A \rangle) = sk_i$, and then the $\mathcal{B}$ returns the corresponding $sk_{i,j}$ to $\mathcal{A}_I$. If $g_{i,j} = 1$, $\mathcal{B}$ will abort the process.

Encryption Key Replacement Query: For the query, the adversary $\mathcal{A}_I$ can randomly choose a new encryption key PK'_{serv_t} to replace the original encryption key PK_{serv_t} corresponding to $serv_t$.

Encryption Query: When the adversary $\mathcal{A}_I$ makes an encryption query with $\langle m, ID_{ES}^A, ID_{ES}^B, PID_{i,j}^A, serv_\alpha, T_i \rangle$ (assume that $\mathcal{A}_I$ has already executed extract encryption key query with $\langle serv_\alpha \rangle$), $\mathcal{B}$ queries the PK_{serv_α} corresponding

to $serv_\alpha$. If $c_\alpha = 1$, $\mathcal{B}$ will abort the process; otherwise, $\mathcal{B}$ obtains $sk_{i,j}^A$ and PK_{serv_α} by querying Map_{ssk} and Map_{EK} respectively. Then, $\mathcal{B}$ runs the *Sign* algorithm and *Encrypt* algorithm to generate the final encryption message $\delta = (c = (s, Y_i), W, PID_{i,j}^A, PK_{i,j}^A, T_i)$, final the $\mathcal{B}$ sends δ to $\mathcal{A}_I$.

Decryption Query: When the adversary $\mathcal{A}_I$ makes a decryption query with $\langle \delta, ID_{ES}^A, ID_{ES}^B, serv_\alpha \rangle$ (assume that $\mathcal{A}_I$ has already executed extract encryption key query with $\langle serv_\alpha \rangle$), $\mathcal{B}$ queries the PK_{serv_α} corresponding to $serv_\alpha$. If $c_\alpha = 0$, $\mathcal{B}$ obtains sk_{serv_α} by querying Map_{ssk} . Then $\mathcal{B}$ runs the *Decrypt* algorithm to obtain m , final the $\mathcal{B}$ returns the m to $\mathcal{A}_I$, if the δ is invalid, the $\mathcal{B}$ will return $\perp$. If $c_\alpha = 1$, $\mathcal{B}$ obtains r_H, r_{H_3} by querying Map_{H_3} , $\mathcal{B}$ obtains r_{H_4} by querying Map_{H_4} . Then the $\mathcal{B}$ calculates $m||\sigma = s \oplus r_{H_4}$ and obtains r_{H_5} by querying Map_{H_5} , if $PK_{i,j}^A = \sigma(r_{H_5} \cdot W + (r_H + r_{H_3}) \cdot U)$, $\mathcal{B}$ returns the m to $\mathcal{A}_I$. If δ is invalid, the $\mathcal{B}$ will return $\perp$. If there is no value for $\langle serv_\alpha \rangle$ in Map_{EK} , it means that encryption key was replaced. Then the $\mathcal{B}$ obtains r'_H, r'_{H_3} by querying Map_{H_3} and obtains r'_{H_4} by querying Map_{H_4} . Next, the $\mathcal{B}$ calculates $m||\sigma = s \oplus r'_{H_4}$ and obtains r'_{H_5} by querying Map_{H_5} , if $PK_{i,j}^A = \sigma(r'_{H_5} \cdot W + (r'_H + r'_{H_3}) \cdot U)$, $\mathcal{B}$ returns the m to $\mathcal{A}_I$. Otherwise, $\mathcal{B}$ outputs $\perp$.

Challenge: Adversary $\mathcal{A}_I$ outputs a challenged pseudonym $PID_{e,f}^A$, a challenged service $serv_\alpha$ and two plaintext m_0, m_1 . $\mathcal{B}$ executes extract encryption key query on $serv_\alpha$, and obtains the value PK_{serv_α} . If $c_\alpha = 0$, $\mathcal{B}$ aborts the process. Otherwise, $\mathcal{B}$ chooses random number $a, b, w \in \mathbb{Z}_q^*$, sets $U = a \cdot P$ and $PK_{head}^B = b \cdot P$. Then the $\mathcal{B}$ obtains r_H, r_{H_3} by querying Map_{H_3} and obtains r_{H_4}, r_{H_5} by querying Map_{H_4}, Map_{H_5} respectively. Next, $\mathcal{B}$ computes $\sigma = sk_{e,f}^A \cdot (r_{H_5} \cdot w + (r_H + r_{H_3}) \cdot a)^{-1}$, $s = r_{H_4} \oplus (m_d||\sigma)$, $Y_i^A = a \cdot (PK_{serv_\alpha} + r_H \cdot PK_{head}^B)$. Finally, $\mathcal{B}$ sends the challenged message $\delta = (c = (s, Y_i^A), W, PID_{e,f}^A, PK_{e,f}^A, T_i)$ to $\mathcal{A}_I$.

After Adversary $\mathcal{A}_I$ executes the above-mentioned query of the probability polynomial time and outputs the guess value $d' \in \{0, 1\}$. If $d' = d$, $\mathcal{B}$ outputs $abP = \frac{1}{r_H}(Y_i^A - r_{know}U)$ as the valid solution of the ECDHP, where $Y_i^A = a(PK_{serv_\alpha} + r_H \cdot PK_{head}^B) = (r_{know} + r_H \cdot b) \cdot U$, $U = aP$, $r_H = H_3(ID_{ES}^B, PK_{ES}^B, PK_{ES}^A)$. Otherwise, $\mathcal{B}$ did not solve the ECDHP.

If the simulator $\mathcal{B}$ does not abort during the simulation, and the adversary $\mathcal{A}_I$ broke the confidentiality of the scheme with a non-negligible probability ε_I , then $\mathcal{B}$ outputs the valid solution of ECDHP. Assume that ε'_I denotes $\mathcal{A}_I$ did not perform extract decryption key query for the challenged service, then $Pr[\varepsilon'_I] = 1 - \frac{q_{dsk}}{2^\lambda}$; ε''_I denotes $\mathcal{B}$ did not abort during the query phase, then $Pr[\varepsilon''_I] = (1 - \tau)^{q_{enc} + q_{dsk}}$; ε'''_I denotes $\mathcal{B}$ did not abort during the challenge phase, then $Pr[\varepsilon'''_I] = \tau$.

The probability that $\mathcal{B}$ does not abort in the entire simulation process is $Pr[\varepsilon'_I \wedge \varepsilon''_I \wedge \varepsilon'''_I] = (1 - \frac{q_{dsk}}{2^\lambda})(1 - \tau)^{q_{enc} + q_{dsk}}\tau$, where $\tau = \frac{1}{q_{enc} + q_{dsk} + 1}$. When $q_{enc} + q_{dsk}$ is large enough, $(1 - \frac{1}{q_{enc} + q_{dsk} + 1})^{q_{enc} + q_{dsk} + 1}$ tends to e^{-1} . Therefore, the probability that $\mathcal{B}$ does not abort during the simulation is at least $(1 - \frac{q_{dsk}}{2^\lambda}) \frac{1}{e(q_{enc} + q_{dsk})}$.

In summary, if simulator $\mathcal{B}$ does not abort during the simulation process, and adversary $\mathcal{A}_I$ break the confidentiality of the scheme with a non-negligible probability ε_I , then the

simulator $\mathcal{B}$ with probabilistic polynomial time outputs the valid solution of ECDHP with a non-negligible probability no less than $(1 - \frac{q_{dsk}}{2^\lambda}) \frac{\varepsilon_I}{e(q_{enc} + q_{dsk})}$.

2) Unforgeability

Theorem 2: In the random oracle model, if an adversary $\mathcal{A}_{II}$ with probabilistic polynomial time executes *Game 2* and wins the game with a non-negligible probability ε_{II} , then the simulator $\mathcal{B}$ with probabilistic polynomial time can solve the ECDLP problem with a non-negligible probability no less than $(1 - \frac{q_{ssk}}{2^\lambda}) \frac{\varepsilon_{II}}{e(q_s + q_{ssk})}$, where q_{ssk} denotes the maximum times of extracting signature key queries, q_s denotes the maximum times of sign queries.

Proof: If there is an adversary $\mathcal{A}_{II}$ that can break our proposed scheme with non-negligible probability ε_{II} , then we can construct a simulator $\mathcal{B}$ based on $\mathcal{A}_{II}$, and the $\mathcal{B}$ can solve the ECDLP run by $\mathcal{A}_{II}$ as a subroutine with a non-negligible probability. Given a group $\mathbb{G}$ and an ECDLP instance $\{P, Q = bP | b \in \mathbb{Z}_q^*, P \in \mathbb{G}, Q \in \mathbb{G}\}$, $\mathcal{B}$ simulates oracles queried by $\mathcal{A}_{II}$ as follows.

Adversary $\mathcal{A}_{II}$ executes setup, H_1, H_2, H_3, H_4, H_5 query, extract verification key query and extract signature key query of the random oracle in *Theorem 1*.

Verification Key Replacement Query: For the query, the adversary $\mathcal{A}_{II}$ can randomly choose a new verification key $PK_{i,j}^{A'}$ to replace the original verification key $PK_{i,j}^A$ corresponding to $PID_{i,j}^A$.

Sign-Query: For the query, $\mathcal{B}$ presets a map Map_{sig} , and the Map_{sig} is empty at the beginning. When the adversary $\mathcal{A}_{II}$ makes a signing query with $\langle m, PID_{e,f}^A \rangle$ (assume that $\mathcal{A}_{II}$ has already executed extract verification key query with $\langle PID_{e,f}^A \rangle$), $\mathcal{B}$ first queries the value $sk_{e,f}^A$ of Map_{ssk} . If $g_{e,f} = 1$, $\mathcal{B}$ aborts the process. Otherwise, $\mathcal{B}$ chooses some random numbers $w, u \in \mathbb{Z}_q^*$, then $\mathcal{B}$ obtains r_H, r_{H_3} by executing H_3 -query and obtains r_{H_5} by executing H_5 -query. Then, $\mathcal{B}$ runs the *Sign* algorithm and *Encrypt* algorithm to generate the final encryption message $\delta = (c = (s, Y_i), W, PID_{e,f}^A, PK_{e,f}^A, T_i)$, final the $\mathcal{B}$ sends δ to $\mathcal{A}_{II}$.

Verify Signature Query: When the adversary $\mathcal{A}_{II}$ makes a verify signature query with $\delta = (c = (s, Y_i), W, PID_{e,f}^A, T_i)$ (assume that $\mathcal{A}_{II}$ has already executed extract verification key query with $\langle PID_{e,f}^A \rangle$), $\mathcal{B}$ queries the $PK_{e,f}^A$ corresponding to $PID_{e,f}^A$. If $g_{e,f} = 0$, $\mathcal{B}$ runs the *Decrypt* algorithm to obtain m , final the $\mathcal{B}$ returns the m to $\mathcal{A}_{II}$. If the δ is invalid, the $\mathcal{B}$ will return $\perp$. If $g_{e,f} = 1$, $\mathcal{B}$ obtains r_H, r_{H_3}, r_{H_4} by querying Map_{H_3}, Map_{H_3} and Map_{H_4} respectively. Then the $\mathcal{B}$ calculates $m||\sigma = s \oplus r_{H_4}$ and obtains r_{H_5} by querying Map_{H_5} , if $PK_{e,f}^A = \sigma(r_{H_5} \cdot W + (r_H + r_{H_3}) \cdot U)$, $\mathcal{B}$ returns the m to $\mathcal{A}_{II}$. If δ is invalid, the $\mathcal{B}$ will return $\perp$. If there is no value for $\langle PID_{e,f}^A \rangle$ in Map_{VK} , it means that verification key was replaced. Then the $\mathcal{B}$ obtains r'_H, r'_{H_3}, r'_{H_4} by querying Map_{H_3}, Map_{H_3} and Map_{H_4} respectively. Next, the $\mathcal{B}$ calculates $m||\sigma = s \oplus r'_{H_4}$ and obtains r'_{H_5} by querying Map_{H_5} , if $PK_{e,f}^A = \sigma(r'_{H_5} \cdot W + (r'_H + r'_{H_3}) \cdot U)$, $\mathcal{B}$ returns the m to $\mathcal{A}_{II}$. Otherwise, $\mathcal{B}$ outputs $\perp$.

Forgery: After executes the above-mentioned query of the probability polynomial time, adversary $\mathcal{A}_{II}$ outputs a forged signature σ . If $\mathcal{A}_{II}$ successfully forges the signature and $g_{e,f} =$

1, then $\mathcal{B}$ outputs $b = r_{know2}$ as the valid solution of ECDLP. Otherwise, $\mathcal{B}$ did not solve ECDLP.

Assume that ε'_{II} denotes $\mathcal{A}_{II}$ did not perform extract signature key query for the challenge pseudonym $PID_{e,f}^A$, then $Pr[\varepsilon'_{II}] = 1 - \frac{q_{ssk}}{2^\lambda}$. Assume that ε''_{II} denotes $\mathcal{B}$ did not abort during querying phase, then $Pr[\varepsilon''_{II}] = (1 - \tau)^{q_s + q_{ssk}}$. Assume that ε'''_{II} denotes $\mathcal{B}$ did not abort in the forgery phase, then $Pr[\varepsilon'''_{II}] = \tau$. Hence, the probability that $\mathcal{B}$ does not abort in the entire simulation process is $Pr[\varepsilon'_{II} \wedge \varepsilon''_{II} \wedge \varepsilon'''_{II}] = (1 - \frac{q_{ssk}}{2^\lambda})(1 - \tau)^{q_s + q_{ssk}}\tau$, where $\tau = \frac{1}{q_s + q_{ssk} + 1}$. When $q_s + q_{ssk}$ is large enough, $(1 - \frac{1}{q_s + q_{ssk} + 1})^{q_s + q_{ssk} + 1}$ tends to e^{-1} . Therefore, the probability that $\mathcal{B}$ does not abort during the simulation is at least $(1 - \frac{q_{ssk}}{2^\lambda})\frac{1}{e(q_s + q_{ssk})}$.

In summary, if simulator $\mathcal{B}$ does not abort during the simulation process, and adversary $\mathcal{A}_{II}$ broke the unforgeability of the scheme with a non-negligible probability ε_{II} , then the simulator $\mathcal{B}$ with probabilistic polynomial time outputs the valid solution of ECDLP with a non-negligible probability of no less than $(1 - \frac{q_{ssk}}{2^\lambda})\frac{\varepsilon_{II}}{e(q_s + q_{ssk})}$.

B. Security Analysis

This subsection gives a detailed analysis of various security features that our proposed scheme satisfies.

1) *Message Confidentiality*: According to *Theorem 1*, we know that no polynomial adversary can challenge success if the ECDHP is hard. Before sending a message, the smart device first encrypts the plaintext m by calculating $s = H_4(U, T_i) \oplus (m || \sigma)$. During transmission, messages are in ciphertext, and u, U are stored in the message sender. Through calculation or guessing, malicious network attackers and illegal receivers cannot obtain u or U . In addition, only the legitimate message receiver with the service-based secret key sk_{serv} can obtain U by computing $Y_i^A \cdot sk_{serv}^{-1}$, and further, obtain the plaintext m by calculating $H_4(U', T_i) \oplus s$. Therefore, the scheme can guarantee the confidentiality of the message.

2) *Message Integrity and Authentication*: According to the *Theorem 2*, it can be concluded that if the ECDLP is difficult to solve, then no adversary can forgery a legal signature within a given polynomial time. In addition, as long as the message and signature meet $PK_{i,j}^A = \sigma' \cdot (H_5(m') \cdot W' + (h + h'_3) \cdot U')$, the signature σ' is proven to be valid. Therefore, the scheme can guarantee the integrity and authentication of the message.

3) *Message Anonymity*: In the process of cross-domain authentication, the smart device SD_i^X does not use its real identity when signing and encrypting, but uses pseudonym $PID_{i,j}^X = RID_i^X \oplus H_2(sk_{i,j}^X \cdot P_{pub}^X)$ and $sk_{i,j}^X = H_1(r_j \cdot P_{pub}) + msk^X$, where $r_j \in \mathbb{Z}_q^*$. To obtain the real identity RID_i^X of SD_i^X , the pseudonym $PID_{i,j}^X$ and secret key $sk_{i,j}^X$ of SD_i^X must be obtained. On the one hand, the $sk_{i,j}^X$ is not transmitted on the network; on the other hand, according to the ECDLP, network attackers and message receivers cannot obtain the secret key $sk_{i,j}^X$ through the public key $PK_{i,j}^X$. Therefore, except for the pseudonym owner and TA^X , no other entity can obtain the real identity based on public information.

4) *Un-Linkability*: In our proposed scheme, the smart device SD_i^X uses a pseudonym $PID_{i,j}^X$ to generate a signature. Note

that generating $PID_{i,j}^X$ needs a random number r_j , and each random number is unique and un-linkability. So no adversary can link the two different signatures generated by two different pseudonyms from the same smart device. Therefore, our proposed scheme supports unlinkability.

5) *Traceability*: TA^X can extract the pseudonym $PID_{i,j}^X$ from the message sent by a smart device SD_i^X , and it can calculate the SD_i^X 's real identity RID_i^X through the $PID_{i,j}^X$. If a smart device is found to send an illegal message, the IIoT system will feed the illegal message back to TA^X . TA^X first queries the public key $PK_{i,j}^X$ corresponding to $PID_{i,j}^X$, and then calculates $RID_i^X = H_2(msk^X \cdot PK_{i,j}^X) \oplus PID_{i,j}^X$ to obtain the real identity. Therefore, in our proposed scheme, TA can trace the source of any illegal message.

6) *Identity Revocation*: In our proposed scheme, TA can cooperate with the blockchain to revoke illegal identity. If a smart device SD_i^X is found to publish an illegal message, TA^X first calculates the real identity RID_i^X corresponding to the illegal message and then stops generating the pseudonym $PID_{i,j}^X$. Subsequently, TA packs all the current pseudonyms into $PIDS_i^X$ and forwards the delete request for $PIDS_i^X$ to the blockchain. Finally, the blockchain can delete the corresponding information according to the delete request. If the illegal smart device regenerates a signature, ES can't find valid $PID_{i,j}^X$ in the blockchain, so the signature will be discarded. Therefore, our proposed scheme supports identity revocation.

7) *Resistance to Replay Attack*: In the proposed scheme, the message receiver should check whether the message has expired before verification. Assuming that T_1 represents the timestamp when the message is sent, T_2 represents the timestamp when the message is received, and ΔT represents the maximum delay time of the message. If $T_2 - T_1 \leq \Delta T$, the receiver will further authenticate the message; otherwise, the message will be considered expired, and the receiver will directly discard the message. Therefore, our proposed scheme can withstand replay attack.

8) *Resistance to Modification Attack*: According to the *Theorem 2*, we know that if an attacker modifies any parameter in $\{c, W, PID_{i,j}^A, PK_{i,j}^A, T_i\}$, the message receiver can determine that some parameters were modified by verifying that $PK_{i,j}^{A'} = \sigma' \cdot (H_5(m') \cdot W' + (h + h'_3) \cdot U')$ does not hold. Therefore, the proposed scheme can withstand the modification attack.

9) *Resistance to Impersonation Attack*: Suppose an attacker wants to impersonate a legitimate smart device. In that case, the attacker should have the ability to construct legitimate parameters $\{c, W, PID_{i,j}^A, PK_{i,j}^A, T_i\}$ and needs to ensure that the formula $PK_{i,j}^{A'} = \sigma' \cdot (H_5(m') \cdot W' + (h + h'_3) \cdot U')$ holds. According to the *Theorem 2*, it is impossible for an attacker to generate the above legitimate parameters. Therefore, the proposed scheme can withstand the impersonation attack.

C. Security Comparisons

We compare the security of our proposed scheme with three recently proposed authentication schemes for IIoT. In addition, we use SR-1, SR-2, SR-3, SR-4, SR-5, SR-6, SR-7, SR-8, and SR-9 to denote message confidentiality, message integrity and

TABLE II
SECURITY COMPARISON OF FOUR SCHEMES

	Cui <i>et al.</i> [28]	Shen <i>et al.</i> [19]	Yang <i>et al.</i> [34]	Our proposed
SR-1	✓	✓	×	✓
SR-2	✓	✓	✓	✓
SR-3	✓	✓	✓	✓
SR-4	✓	×	✓	✓
SR-5	✓	✓	✓	✓
SR-6	×	✓	✓	✓
SR-7	×	×	✓	✓
SR-8	✓	✓	✓	✓
SR-9	✓	✓	✓	✓

✓: The requirement is satisfied.

×: The requirement is not satisfied.

authentication, message anonymity, un-linkability, traceability, identity revocation, resistance to replay attack, resistance to modification attack, and resistance to impersonation attack, respectively. The results of the security comparison are shown in Table II.

It is worth noting that in [19], the confidentiality of the message is satisfied only after the message sender encrypts the message using the negotiated key. Still, during the subsequent interaction, the message sender uses this key all the time, so the un-linkability of the message is not satisfied. In [19] and [34], only the anonymity of the message senders is guaranteed, while in [28] and our proposed scheme, the anonymity of both the message senders and the message receivers are guaranteed. Because [28] and our proposed scheme are designed for a multi-receiver scenario, so the sender does not know which specific device the receiver is. However, [19] and [34] are not suited for a multi-receiver cross-domain IIoT scenario, so the sender is supposed to know exactly which device the receiver is and send the message directed.

VII. PERFORMANCE EVALUATION

For a cross-domain IIoT scenario, the computing power of smart devices is limited, but the scenario requires a high level of real-time data. To demonstrate the feasibility of our proposed scheme, we evaluate the performance in terms of three aspects: computation overhead, communication overhead, and query latency.

A. Experimental Settings

1) *Comparison Schemes Setting*: To demonstrate the effectiveness of our proposed scheme, we compare it with three other schemes. To make the comparison more fair and convenient, in all schemes, we set up that all smart devices from different domains cannot communicate with each other directly and must communicate with each other through the edge server. Other corresponding changes are described in detail as follows.

- In Cui et al.'s scheme [28], SD_i^A signs the initial message, ES^B re-encrypts the message, and SD_k^B authenticates the received message.

- In Shen et al.'s scheme [19], SD_i^A and SD_k^B need to negotiate an encryption key before message authentication, and then SD_i^A sends the encrypted message to SD_k^B . So it takes three rounds of interaction between SD_i^A and SD_k^B to complete a message authentication. During these three rounds of interactions, the generation of the initial message and key negotiation is done by smart devices. The message signing, verification, and on-chain querying are all done by edge servers. Note that the encryption algorithm used in our experiments is Advanced Encryption Standard (AES).
- In Yang et al.'s scheme [34], SD_i^A signs the initial message, ES^B queries the necessary information from the blockchain, and SD_k^B authenticates the received message.

2) *Experimental Environment Settings*: To evaluate the performance of the proposed scheme in a real scenario, we use C++ to implement these schemes. In the experiment, the cryptographic tool library we use is Miracl Core [41]. And we choose the BLS12381 type curve, which provides 128-bit security level. In addition, all the common secure hash functions we use in this experiment first convert the input parameters to binary, then hash them using the SHA256 function and finally convert them to the required data type.

In our experiments, we use a PC running Ubuntu 18.04.3 operating system to simulate ES; this PC is equipped with an Intel Core i5-7500 CPU @3.4 GHz and 16 GB of memory. Since the computing power of SD is limited, we use a Raspberry Pi 4 to simulate the SD. This Raspberry Pi 4 runs Debian GNU/Linux 11 operating system and is equipped with a 1.5 GHz CPU and 4 GB of memory.

To facilitate testing the time spent on the corresponding operations in the blockchain, we deploy a blockchain platform. This platform uses the hyperledger fabric [42] and is deployed on nine PCs with the same performance. All nine PCs are running Ubuntu 18.04.3 operating system and have Intel Core i7-11700 CPU @2.50 GHz and 16 GB of memory. Among these nine PCs, one PC acts as an orderer node, and each of the other PCs acts as a BCDA in a certain administrative domain, respectively. We built different channels using different numbers of BCDAs to simulate consortium blockchains built by different numbers of domains. In addition, we use a PC equipped with an Intel Core i5-7500 CPU @3.4 GHz and 16 GB of memory as the client node. The on-chain operations we implement using the Go code.

It is worth noting that we assume that a cross-domain IIoT service has n receivers and the experimental results we show are the average of 200 experiments.

B. Computation Overhead

To compare the computational overhead more comprehensively, we first perform a theoretical analysis of the computational overhead and then compare the computational overhead by analyzing experimental simulation results.

1) *Theoretical Analysis*: To make the theoretical analysis more convenient, we define some notations for some time-consuming operations as follows. Note that according to the

TABLE III
TIME-CONSUMING CRYPTOGRAPHIC OPERATIONS IN THE FOUR SCHEMES

Scheme	Without Pre-processing				With Pre-processing			
	SD_i^A	ES^A	ES^B	SD_k^B	SD_i^A	ES^A	ES^B	SD_k^B
Cui <i>et al.</i> [28]	$10E_1 + 3E_2 + 3B + H$	-	E_1	$9E_1 + 4E_2 + 5B$	-	-	E_1	$9E_1 + 4E_2 + 2B$
Shen <i>et al.</i> [19]	$2nS_1$	$nS_1 + 2nE_2 + nS_2 + 3nB$	$nS_1 + 2nE_2 + nS_2 + 3nB$	$2S_1$	nS_1	$nS_1 + nS_2 + nE_2 + nB$	$nS_1 + nS_2 + nE_2 + nB$	S_1
Yang <i>et al.</i> [34]	$3nS_1$	-	-	$3S_1$	-	-	-	$3S_1$
Our proposed	$4S_1$	-	-	$4S_1$	-	-	-	$4S_1$

experimental setup, a message has n receivers. In addition, “-” in Table III indicates no time-consuming cryptographic operations.

- B : a bilinear pairing operation $e(g_1, g_2)$, where $g_1 \in \mathbb{G}_1$, $g_2 \in \mathbb{G}_2$.
- S_1 : a scalar point multiplication operation $a \cdot P_1$, where $a \in \mathbb{Z}_q^*$ and $P_1 \in \mathbb{G}_1$.
- S_2 : a scalar point multiplication operation $a \cdot P_2$, where $a \in \mathbb{Z}_q^*$ and $P_2 \in \mathbb{G}_2$.
- E_1 : an exponential operation P_1^r , where $r \in \mathbb{Z}_q^*$ and $P_1 \in \mathbb{G}_1$.
- E_2 : an exponential operation P_3^r , where $r \in \mathbb{Z}_q^*$ and $P_3 \in \mathbb{G}_T$.
- H : a hash-to-point operation $H(m) \in \mathbb{G}_1$, where $H(\cdot)$ is a secure hash function and $m \in \{0, 1\}^*$.

In each scheme, specific information on the time-consuming cryptographic operations on each entity is shown in Table III.

When there is no pre-processing in the authentication process, we can see that in scheme [28], most of the time-consuming cryptographic operations are performed by SD_i^A and SD_k^B , and the time-consuming cryptographic operations performed by SD_i^A and SD_k^B are $10E_1 + 3E_2 + 3B + H$ and $9E_1 + 4E_2 + 5B$ respectively. In scheme [19], we can see that the time-consuming cryptographic operations that need to be performed by entities other than the SD_k^B increase with the number of receivers. In scheme [34], the time-consuming cryptographic operation performed by the SD_i^A increases with the number of receivers, which is $3nS_1$. And the time-consuming cryptographic operation performed by the SD_k^B is $3S_1$. In our proposed scheme, the time-consuming cryptographic operations performed by the SD_i^A and SD_k^B are fixed, both being $4S_1$. That is, in our proposed scheme, the time-consuming cryptographic operations performed by SD_i^A and SD_k^B do not vary with the number of receivers.

Many operations can be pre-processed offline before the message is generated. As shown in Table III, with pre-processing, the time-consuming operations of the various schemes are effectively reduced. For example, in scheme [28], the time-consuming cryptographic operations performed by SD_i^A and SD_k^B are reduced by $10E_1 + 3E_2 + 3B + H$ and $(9E_1 + 4E_2 + 5B) - (9E_1 + 4E_2 + 2B) = 3B$, respectively. In scheme [19], the

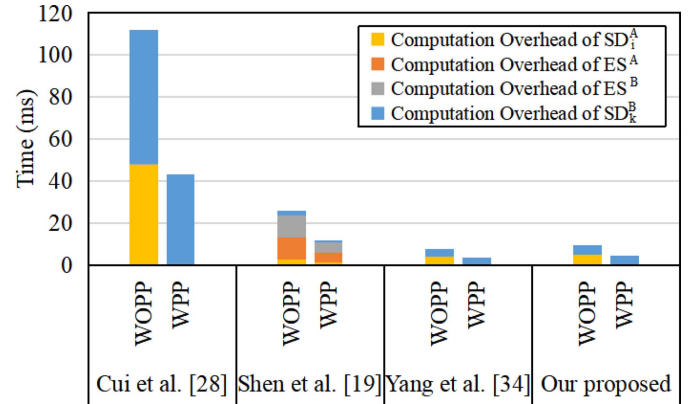


Fig. 4. Comparison of computation overhead when the number of message receivers is 1 (WPP: with pre-processing, WOPP: without pre-processing).

time-consuming cryptographic operations to be performed by SD_i^A , ES^A , ES^B and SD_k^B are reduced by nS_1 , $(nE_2 + 2nB)$, $(nE_2 + 2nB)$ and S_1 respectively. Similarly, we find that in scheme [34], the time-consuming cryptographic operations to be performed by SD_i^A are reduced by $3nS_1$. In our proposed scheme, the time-consuming operations performed by SD_i^A are reduced by $4S_1$. It is worth noting that the time-consuming operation required for SD_k^B in our proposed scheme is $4S_1 - 3S_1 = 1S_1$ more than that required for SD_k^B in [34], because our proposed scheme guarantees data confidentiality while [34] does not, and the $1S_1$ is required for decryption operation in our proposed scheme. Data confidentiality is achieved using fewer cryptographic operations, which we consider to be an effective trade-off between security and efficiency.

2) *Simulation Experimental Results*: To illustrate that our proposed scheme is lightweight more concretely, we implement each scheme with and without pre-processing according to the experimental setting. When the number of receivers is 1, the experimental results are shown in Table IV and Fig. 4. Note that all cryptographic operations (e.g., integer addition and multiplication) are included in the simulation experiments. Therefore, in Table IV, when schemes support pre-processing, computational overhead exists for the SD_i^A in the scheme [34]

TABLE IV
SIMULATION EXPERIMENTAL RESULTS OF THE FOUR SCHEMES (MS)

Scheme	Without Pre-processing				With Pre-processing			
	SD_i^A	ES^A	ES^B	SD_k^B	SD_i^A	ES^A	ES^B	SD_k^B
Cui <i>et al.</i> [28]	47.746	-	0.278	63.971	0.029	-	0.277	42.654
Shen <i>et al.</i> [19]	2.523	10.394	10.387	2.496	1.266	4.664	4.660	1.263
Yang <i>et al.</i> [34]	3.900	-	-	3.552	0.035	-	-	3.235
Our proposed	4.790	-	-	4.597	0.064	-	-	4.516

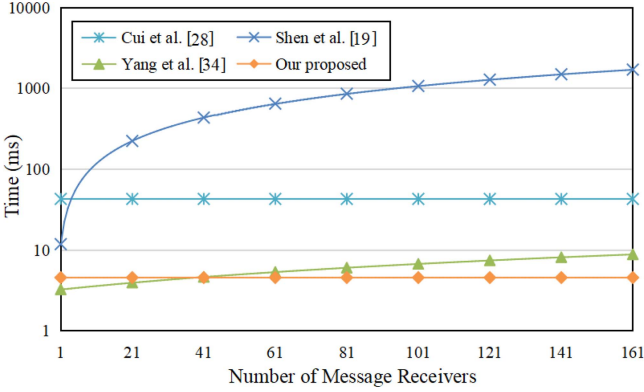


Fig. 5. Authentication time consumption for different number of message receivers (with pre-processing).

and our proposed scheme. In addition, Fig. 5 compares the total time required for authentication in each scheme as the number of message receivers increases. The “-” in Table IV indicates no computational overhead.

Combining Table IV and Fig. 4, we can see that when the number of receivers is 1, the total time cost to complete an authentication process in [28] without pre-processing is $47.746 + 0.278 + 63.971 = 111.995$ ms, which is the highest among the four schemes. The reason is that resource-constrained smart devices perform many time-consuming cryptographic operations. Similarly, the scheme [19] contains many time-consuming cryptographic operations, and the total time required to complete an authentication is $2.523 + 10.394 + 10.387 + 2.496 = 25.800$ ms. Fig. 4 shows that the total time required to complete an authentication between [34] and our proposed scheme is very low, where [34] takes $3.900 + 3.552 = 7.452$ ms, which is the lowest time cost among the four schemes. In our proposed scheme, the time required is $(4.790 + 4.597) - 7.452 = 1.935$ ms more than that needed by [34], but this gap will be effectively reduced with pre-processing.

From Table IV, we find that the time cost of our scheme with pre-processing is saved about $((4.790 + 4.597) - (0.064 + 4.516)) / (4.790 + 4.597) \approx 51.2\%$ relative to the case without pre-processing. We can see that with pre-processing, the time spent in our proposed scheme is about $(0.064 + 4.516) / (0.029 + 0.277 + 42.654) \approx 10.7\%$ of [28] and $(0.064 + 4.516) / (1.266 + 4.664 + 4.660 + 1.263) \approx 38.6\%$ of [19]. In addition, with pre-processing,

the time cost of our proposed scheme is $(0.064 + 4.516) - (0.035 + 3.235) = 1.310$ ms more than [34]. The reason is that our proposed scheme achieves data confidentiality and requires a decryption operation while [34] does not.

In Fig. 5, we can see that the time spent in [19] grows fastest as the number of message receivers increases. The time spent in [28] and our proposed scheme does not increase with the number of message receivers, which is because both schemes are designed for multi-receiver IIoT scenarios. In addition, we find that the time consumption of [34] and our proposed scheme have been at a low level. However, when the number of message receivers exceeds 38, the time to be consumed by [34] will exceed that of our proposed scheme. Therefore, the computational overhead of our proposed scheme is better than other related schemes when the number of receivers exceeds 38.

In summary, our proposed scheme is lightweight and suitable for multi-receiver cross-domain IIoT.

C. Communication Overhead

According to the experimental setting, to compare the communication overhead of each scheme, we first record the interactions between devices throughout the process from signature generation to signature authentication, then compute the size of communication packets. In the cryptographic tool library we use, the occupied space size for elements in $\mathbb{Z}_q^*$ is 48 bytes, and the occupied space size for elements in $\mathbb{G}_1$ is 97 bytes. We set the size of the message is 22 bytes and set the size of the timestamp is 16 bytes. In addition, we let I_{SD-ES} , I_{ES-ES} , I_{ES-BC} denote the interaction between the smart device and the edge server, the interaction between the edge server and the edge server, the interaction between the edge server and the blockchain respectively.

In our proposed scheme, the smart device SD_i^A sends data $\delta = (c, W, PID_{i,j}^A, T_i)$ to ES^A , where the size of this data is $(22 + 48 + 97 + 97 + 32 + 16) = 312$ bytes. The ES^A then forwards the received data to the ES^B , which is 312 bytes. Once the ES^B receives the data, it will send $PID_{i,j}^A$ of size 32 bytes to $BCDA^B$ and then get the $PK_{i,j}^A$ corresponding to the $PID_{i,j}^A$. Finally, the ES^B sends data $(c, W, PID_{i,j}^A, T_i, PK_{i,j}^A)$ to SD_k^B , where the size of this data is $(312 + 97) = 409$ bytes. In addition, there are n receivers for an IIoT service. Therefore, in our proposed scheme, the total communication packet size is $(312 + 312 + 32 + 97) + 409n = 753 + 409n$ bytes and the total interaction is $(n + 1)I_{SD-ES} + I_{ES-ES} + 2I_{ES-BC}$.

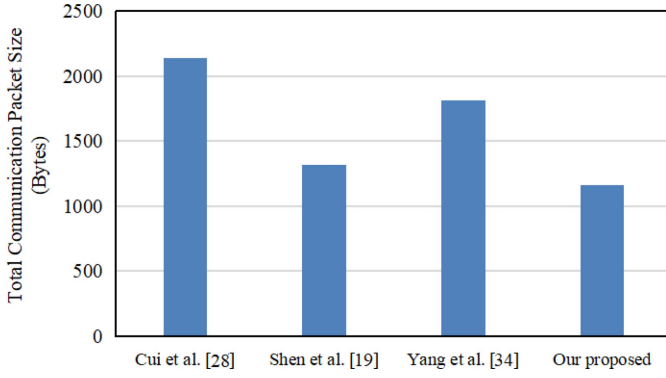


Fig. 6. Comparison of total communication packet size when the number of message receivers is 1.

We compute the communication overhead of other schemes using the same method. In Cui et al.'s scheme [28], the total communication packet size is about $1444 + 698n$ bytes, and the total interaction is $(n+1)I_{SD-ES} + I_{ES-ES}$. In Shen et al.'s scheme [19], the total communication packet size is about $1318n$ bytes, and the total interaction is $6nI_{SD-ES} + 3nI_{ES-ES} + 4nI_{ES-BC}$. In Yang et al.'s scheme [34], the total communication packet size is about $1816n$ bytes, and the total interaction is $2nI_{SD-ES} + nI_{ES-ES} + 2nI_{ES-BC}$. First, we find that [28] has the lowest interactions since the authentication process of this scheme does not involve the blockchain. Second, we find that the total interactions of [34] and [19] are more than the total interactions of our proposed scheme because our proposed scheme leverages service-based ideas and is suitable for the multi-receiver scenario. In our proposed scheme, completing an authentication process only needs one signature, and the edge server only needs to submit one query request to the blockchain. In contrast, the other two related schemes require n signatures and more interactions between entities during the authentication processing (e.g., the edge server needs to submit n query requests to the blockchain). In addition, the comparison of the total communication packet size of the four schemes when the message receiver is 1 is shown in Fig. 6.

From Fig. 6, we can see that our proposed scheme has the smallest total communication packet size among the four schemes. Specifically, the total communication packet size of our proposed scheme is approximately 980 bytes less than that of [28], approximately 156 bytes less than that of [19], and approximately 654 bytes less than that of [34]. In addition, as the number of message receivers increases, the communication overhead advantage of our proposed scheme will become more significant. There are three reasons for this situation. The first reason is that the data transferred per interaction between the entities of our proposed scheme is short, and the second reason is that the total number of interactions between the entities of our proposed scheme is low. The last reason is that our proposed scheme is designed for the multi-receiver cross-domain IIoT scenario. In summary, our proposed scheme has a low communication overhead and is suitable for IIoT environments with high real-time data requirements.

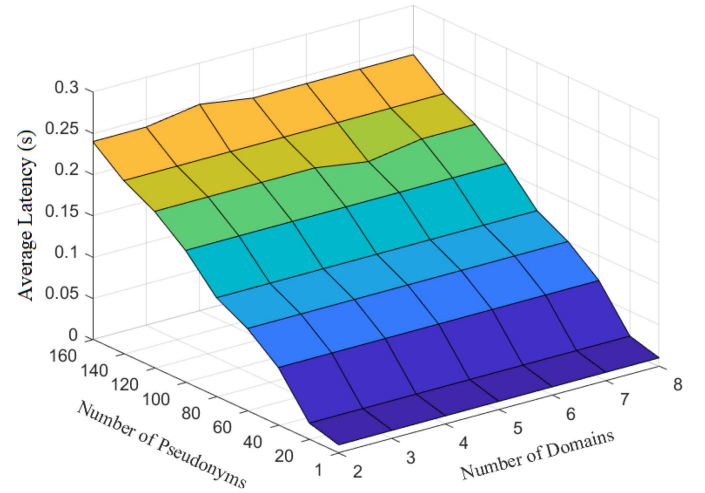


Fig. 7. Query latency.

D. Query Latency

In the process of message authentication, [28], [34], and our proposed scheme need to interact with the blockchain, where these interactions are mainly to perform query operations. Specifically, the ES submits a pseudonym to the blockchain; the blockchain performs the query operation and returns the information corresponding to that pseudonym. Therefore, we evaluate the on-chain query latency based on the blockchain platform settings mentioned in Section VII-A. The results of the experiment are shown in Fig. 7.

From Fig. 7, we can see that when the number of pseudonyms in a batch query is constant, the average latency does not change significantly with the number of domains. For example, regardless of the number of administrative domains, the average latency is about 0.20 s when the number of pseudonyms for a batch query is 20. This is because the query operation is mainly performed in the local ledger. Therefore, the number of domains does not affect the query latency. In addition, we can see that the average query latency rises when the number of pseudonyms in a batch query increases. For example, when the number of pseudonyms for a batch query is 1, the average latency is about 0.01 s. And when the number of pseudonyms in the batch query is 160, the average latency is about 0.24 s.

Insight: To meet the IIoT requirements for real-time data, we should reduce the number of interactions with the blockchain. By analyzing the communication overhead, we find that for an IIoT service, our proposed scheme queries the blockchain only once during the authentication process, regardless of the number of receivers. However, in [28] and [34], the number of queries to the blockchain increases with the number of receivers. Therefore, our proposed scheme is applicable to multi-receiver cross-domain IIoT scenarios.

VIII. CONCLUSION

In this paper, we propose a blockchain-based lightweight message authentication scheme for multi-receiver cross-domain IIoT. The main goal of this scheme is to satisfy the security

and efficiency requirements in cross-domain IIoT. Specifically, we first design a lightweight edge-assisted cross-domain authentication framework using blockchain, and then design a lightweight message authentication algorithm. Detailed security proofs and analysis demonstrate that the proposed scheme can resist various attacks. In addition, a comparison with related schemes shows that our proposed scheme is lightweight in terms of computational and communication overheads and suitable for multi-receiver cross-domain IIoT scenarios. In the future, we will design secure and efficient message authentication schemes for mobile smart devices in IIoT.

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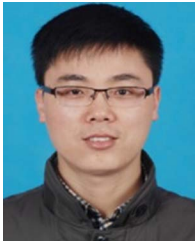
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Fengqun Wang is currently working toward the PhD degree with the School of Computer Science and Technology, Anhui University, Hefei, China. His research interests include IoT security, blockchain, and applied cryptography.



Jie Cui (Senior Member, IEEE) received the PhD degree from the University of Science and Technology of China, in 2012. He is currently a professor and PhD supervisor with the School of Computer Science and Technology, Anhui University. His current research interests include applied cryptography, IoT security, vehicular ad hoc network, cloud computing security, and software-defined networking (SDN). He has more than 150 scientific publications in reputable journals (e.g., the *IEEE Transactions on Dependable and Secure Computing*, *IEEE Transactions on Information Forensics and Security*, *IEEE Journal on Selected Areas in Communications*, *IEEE Transactions on Mobile Computing*, *IEEE Transactions on Parallel and Distributed Systems*, *IEEE Transactions on Computers*, *IEEE Transactions on Vehicular Technology*, *IEEE Transactions on Intelligent Transportation Systems*, *IEEE Transactions on Network and Service Management*, *IEEE Transactions on Industrial Informatics*, *IEEE Transactions on Industrial Electronics*, *IEEE Transactions on Cloud Computing* and *IEEE Transactions on Multimedia*), academic books and international conferences.



Qingyang Zhang received the BEng and PhD degrees in computer science from Anhui University, in 2021. He is currently a lecture with the School of Computer Science and Technology, Anhui University. His research interests include edge computing, computer systems, and security.



Debiao He received the PhD degree in applied mathematics from the School of Mathematics and Statistics, Wuhan University, Wuhan, China, in 2009. He is currently a professor with the School of Cyber Science and Engineering, Wuhan University, Wuhan, China. His main research interests include cryptography and information security, in particular, cryptographic protocols. He has published more than 100 research papers in refereed international journals and conferences, such as the *IEEE Transactions on Dependable and Secure Computing*, *IEEE Transactions on Information Security and Forensic*, and *Usenix Security Symposium*. He is the recipient of the 2018 IEEE Systems Journal Best Paper Award and the 2019 IET Information Security Best Paper Award. His work has been cited more than 10,000 times with Google Scholar. He is in the Editorial Board of several international journals, such as the *Journal of Information Security and Applications*, *Frontiers of Computer Science*, and *Human-centric Computing & Information Sciences*.



Chengjie Gu received the PhD degree from the Nanjing University of Posts and Telecommunications, in 2012. From 2012 to 2017, he was an innovation team leader with the 38th Research Institute of CETC and conducted research and development in the communication and networking sector. Currently, he is a president of Security Research Institute, New H3C Group. He is also studying for postdoctoral fellowship with the USTC. He is a high-level innovation leader of Anhui province and a cybersecurity expert of Zhejiang province in China. His research interests include network security and trusted network architecture, etc.



Hong Zhong received the PhD degree in computer science from the University of Science and Technology of China, in 2005. She is currently a professor and PhD supervisor with the School of Computer Science and Technology, Anhui University. Her research interests include applied cryptography, IoT security, vehicular ad hoc network, cloud computing security, and software-defined networking (SDN). She has more than 200 scientific publications in reputable journals (e.g., the *IEEE Journal on Selected Areas in Communications*, *IEEE Transactions on Parallel and Distributed Systems*, *IEEE Transactions on Mobile Computing*, *IEEE Transactions on Dependable and Secure Computing*, *IEEE Transactions on Information Forensics and Security*, *IEEE Transactions on Intelligent Transportation Systems*, *IEEE Transactions on Multimedia*, *IEEE Transactions on Vehicular Technology*, *IEEE Transactions on Network and Service Management*, *IEEE Transactions on Cloud Computing*, *IEEE Transactions on Industrial Informatics*, *IEEE Transactions on Industrial Electronics* and *IEEE Transactions on Big Data*), academic books and international conferences.